DISTRIBUTED NETWORK SECURITY MONITORING AND MATHEMATICAL THREAT MODELING

A Security Operations Study

1. Overview

A distributed Network Security Monitoring (NSM) and mathematical threat detection framework was built to evaluate the operational visibility of multi-tiered anomaly models against low-observable, distributed data exfiltration signatures. First, a bare-metal hardware cluster architecture is established across isolated nodes to generate and capture high-fidelity network telemetry. Utilizing signal processing principles, a parametric synthetic baseline incorporating long-range temporal dependencies ($1/f$ pink noise) and stochastic jitter is developed to model typical research and development workflows. Finally, three distinct mathematical detection layers: localized attack-likelihood mappings, lagged signature cross-correlations, and wavelet-coherence operators—are implemented and validated. This study demonstrates how distributed, sub-threshold adversarial signals can be isolated from complex background noise without relying on traditional signature-based detection mechanisms.

2. Experimental Setup

To validate the multi-tiered detection frameworks and baseline models under realistic operational conditions, a dedicated, bare-metal distributed Network Security Monitoring (NSM) laboratory testbed was engineered. The environment establishes an isolated, multi-node compute environment designed to generate, capture, and analyze high-fidelity network telemetry without external network pollution. The physical topology incorporates an orchestration and data collection node, a designated traffic injection node, four independent sensor nodes simulating localized user workloads, and a high-performance management host providing centralized storage and telemetry verification.

2.1 Hardware Layer

The physical architecture was constructed using seven discrete, bare-metal compute nodes operating inside a segregated and isolated LAN. Telemetry orchestration and packet collection were performed by a designated collector node, while simulated network conditions and threat actions were driven by an injector node; both nodes were deployed on Raspberry Pi 5 (RP5) single-board computers configured with 4 GB of low-power LPDDR5 RAM. Distributed user activity and monitoring targets were simulated across four independent sensor nodes, deployed on RP5 hardware platforms configured with 8 GB of LPDDR5 RAM.

To eliminate storage host I/O bottlenecks and CPU contention during intensive telemetry replays, a fifth 8 GB RP5 node was integrated into the cluster as a dedicated storage fabric layer. SanDisk 128 GB Extreme microSDXC UHS-I flash memory cards were used as local storage for each node. Centralized data processing and analytical validation were managed by a management host built around an AMD Ryzen 7 7700X desktop processor featuring 8 physical cores (16 threads) operating at a base-to-boost frequency range of 4.5–5.4 GHz and equipped with 32 GB of DDR5 dual-channel memory. High-throughput, low-latency node interconnection was achieved by connecting all physical interfaces via copper media to an

enterprise-grade managed L3 Gigabit Ethernet switch.

2.2 Software Layer

The injector, collector, sensor, and storage nodes ran PiOS with Linux kernel version 6.12, while the management host operated on Ubuntu 25.10 (AMD64). To facilitate a shared cluster filesystem and completely isolate tracking data, the storage node was configured to exclusively export an NFS share, mapping a central directory across all nodes. The collector, injector, storage, and sensor nodes were configured in a headless deployment. Network isolation was enforced via VLAN segmentation and dedicated firewall policies isolated the lab traffic.

The entire network security monitoring fabric, data aggregation routines, parametric baseline algorithms, and threat telemetry processing layers were implemented using Fortran 90. This single-language architecture provided direct auditability and independent verification of detection model implementations, allowing explicit, low-level control over raw array operations and floating-point execution without relying on high-level black-box interpreter dependencies or external scripting frameworks.[1]

3. Foundational Monitoring Fabric

To establish the foundational monitoring fabric of the NSM lab, the sensors maintained per-node heartbeat continuity and published telemetry records to a common share exported by the storage node. This architecture enabled centralized ingestion and aggregation by the collector. The injector, configured as an iperf3 server, provided the endpoint for controlled, artificially generated network traffic from each sensor. A master Fortran 90 orchestration routine executed on the collector synchronized the collection sequence across each sensor node and initiated the scheduled baseline traffic events through iperf3. For validation, telemetry was collected over 140 s using a sensor sampling window of 5 seconds, yielding a total of 28 contiguous samples. The sensors were configured to generate traffic with the injector over the following intervals:

Sensor	**Direction**	**Start Time (s)**	**Duration (s)**
1	Upload	5	10
1	Download	20	10
2	Upload	35	10
2	Download	50	10
3	Upload	65	10
3	Download	80	10
4	Upload	95	10
4	Download	110	10

Table 3.1: Scheduled traffic events for the final Phase 1 validation run.

The results are shown in Fig. 3.1. As seen in the figure, sequential, non-overlapping iperf3 upload and download events were generated between each sensor and the injector, while the collector remained dedicated to centralized orchestration and aggregation. The upper and lower panels show transmitted and received byte deltas, respectively, during each sampling window.[2] The distinct separation in UL/DL spikes for sensors 1–4 confirms that each sensor successfully maintained heartbeat continuity, recorded telemetry locally, and published measurements to the designated storage fabric for centralized collection.

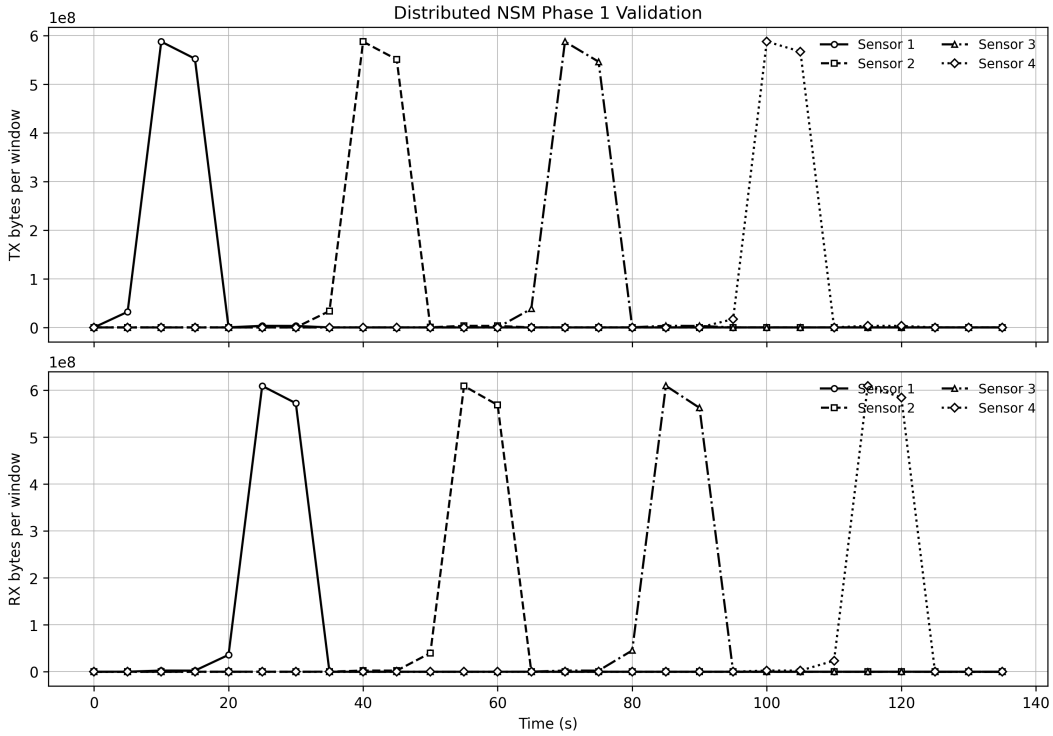


Figure 3.1: Distributed NSM validation. Top: transmitted byte deltas. Bottom: received byte deltas. Sequential `iperf3` events produced distinct, non-overlapping responses for sensors 1–4.

4. Baseline Generation and Detection Framework

To isolate distributed data exfiltration from background network traffic, an analytical framework was modeled using basic principles signal processing. The architecture was split into an injection-validation layer and a mathematical detection layer. First, a background framework was established to coarsely define routine, daily network traffic. the coarse daily traffic was subsequently modulated by long-range correlated noise processes and uncorrelated high-frequency variations to approximate a realistic background network environment. Second, a multilayer mathematical processing engine evaluated the captured telemetry across independent metrics. This framework progressed from single-node localized likelihood functions to distributed lag-correlation matrices, and ultimately to multi-scale wavelet-coherence operators, isolating weak, sub-threshold adversarial footprints from the surrounding environmental noise.

4.1 Baseline Construction

Lacking a realistic enterprise baseline, a synthetic baseline was constructed by partitioning a 24-hour work day into a small number of operational classes that were intended to represent distinct modes of expected R&D activity: overnight quiet, startup/retrieval, active work, wrap-up, and evening baseline mode. For each sensor and baseline class, nominal transmitted and received traffic levels were defined. These values were used to construct the logic-table baseline B_t, which provides the large-scale daily structure of the synthetic telemetry. Subsequently, remaining stochastic and event-driven terms were then added to produce a nontrivial baseline signal with realistic drift, jitter, machine-generated periodic traffic, and human workflow bursts. The baseline classes are shown in Table (4.1)

Class ID	Name	Start Time	End Time
1	Overnight Quiet	00:00	06:00
2	Startup/Retrieval	06:00	09:00
3	Active Work	09:00	17:00
4	Wrap-up	17:00	19:00
5	Evening Baseline	19:00	24:00

Table 4.1: Baseline class schedule for a standard 24-hour operational cycle.

Based upon the baseline class schedule, the baseline network activity was generated synthetically on a 30-second time grid using a rule-based parametric model. Each window was assigned to a time-of-day operating class. For each sensor, separate transmitted and received baselines were constructed from class-dependent nominal byte counts per telemetry window. For either direction, the logic-table baseline was modeled as

$$B(t) = \sum_{i=1}^{n} L_i(t), \tag{4.1}$$

where

$$L_i(t) = \begin{cases} X_i, & t \in W_i, \\ 0, & \text{otherwise.} \end{cases} \tag{4.2}$$

Here, $W_i = [t_i, t_{i+1})$ is the time window associated with the i-th baseline class, and X_i is the nominal transmitted or received byte count assigned to that class. Since the class windows form a partition of the 24-hour timeline, only one $L_i(t)$ contributes at any time. The final synthetic baseline for each sensor and traffic direction was modeled as a Gaussian-smoothed traffic signal,

$$Y(t) = \int_{-\infty}^{\infty} S_{\text{raw}}(\tau)\, G_\sigma(t-\tau)\, d\tau + J(t), \tag{4.3}$$

where $S_{\text{raw}}(t)$ is the unsmoothed synthetic traffic signal, $G_\sigma(t)$ is the Gaussian smoothing kernel, and $J(t)$ is the jitter (explained below). The Gaussian kernel is

$$G_\sigma(t) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{t^2}{2\sigma^2}\right). \tag{4.4}$$

Jitter, $J(t)$, was introduced to simulate high-frequency, uncorrelated variation including, but not limited to, hardware interrupts, thermal noise, and clock skew. The jitter is modeled as additive white Gaussian noise scaled by the logic-table baseline. The jitter is modeled as a normalized stochastic operator. It is generated independently of the baseline logic and then scaled by $B(t)$. The jitter is added to the smoothed $S_{\text{raw}}(t)$ to prevent the smoothing operation (here the convolution integral) from suppressing short-timescale variation.[3] The jitter is given by

$$J(t) = B(t)kZ(t), \tag{4.5}$$

where k is the jitter coefficient and $Z(t)$ is a normalized white Gaussian process. In the implementation, independent samples are generated at each telemetry time index using the Box–Muller transform,

$$Z(t) = \sqrt{-2\ln(U_1(t))}\cos\left(2\pi U_2(t)\right), \tag{4.6}$$

$$U_1(t), U_2(t) \sim \text{Uniform}(0, 1]. \tag{4.7}$$

Here, $U_1(t)$ and $U_2(t)$ denote independent random draws at each time sample, not single functions drawn once over the full interval. Thus, for each time t,

$$Z(t) \sim \mathcal{N}(0, 1), \tag{4.8}$$

with probability density

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad z \in \mathbb{R}. \tag{4.9}$$

The raw signal, $S_{\text{raw}}(t)$, was constructed from the previously defined logic-table baseline, stochastic drift, and simulated event traffic:

$$S_{\text{raw}}(t) = B(t) + D(t) + E(t). \tag{4.10}$$

The drift term $D(t)$ models low-frequency correlated variation, and the event term $E(t)$ models machine-generated periodic traffic and human workflow bursts. Together, these terms generate a nontrivial raw signal prior to Gaussian smoothing. The drift term simulates events including buffer contention, TCP window scaling, and multi-scale background activity. The drift, similar to the jitter, is modeled as a normalized stochastic operator, generated independently of the baseline logic and then scaled by $B(t)$. However, the drift is incorporated into $S_{\text{raw}}(t)$ and subsequently smoothed by convolution, as opposed to the jitter, which was added to the smoothed $S_{\text{raw}}(t)$. It is represented by pink noise calculated by filtering white noise in the frequency domain to achieve a $1/f$ power spectral density. The drift is given by

$$D(t) = B(t)dP(t), \tag{4.11}$$

where d is the drift coefficient and $P(t)$ is the normalized pink-noise process. The pink-noise process is generated according to

$$P(t) = \frac{\widetilde{P}(t) - \mu_{\widetilde{P}}}{\sigma_{\widetilde{P}}}, \tag{4.12}$$

$$\widetilde{P}(t) = \mathcal{F}^{-1}\left[\mathcal{F}(W(t))\,\mathcal{H}_p(f)\right], \tag{4.13}$$

$$\mathcal{H}_p(f) = \begin{cases} 0, & f = 0, \\ f^{-1/2}, & f > 0. \end{cases} \tag{4.14}$$

Here, $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the Fourier and inverse Fourier transforms, $W(t)$ is a white-noise process, f is frequency, and $\mathcal{H}_p(f)$ is the pink-noise spectral weighting function. The zero-frequency component is nulled to avoid a DC offset before normalization. Simulated events were introduced to model background laboratory traffic and human workflow-generated traffic. Laboratory traffic includes regular and quasi-regular traffic such as lab equipment

check-ins, status pings, scheduled machine chatter, and instrumentation updates. Human workflow-generated traffic is irregular and clustered, and includes logins, file transfers, command bursts, interactive work, browser actions, and short task-driven bursts. The simulated event term is modeled as

$$E(t) = \sum_{j=1}^{K} M_j \, \Pi(t; t_j, D_j), \tag{4.15}$$

$$\Pi(t; t_j, D_j) = \begin{cases} 1, & t_j \le t < t_j + D_j, \\ 0, & \text{otherwise.} \end{cases} \tag{4.16}$$

Here, K is the number of human workflow bursts, t_j is the start time of burst j, D_j is the burst duration, and M_j is the burst magnitude. The number of bursts is drawn from a Poisson distribution,

$$K \sim \text{Poisson}(\lambda), \tag{4.17}$$

$$P(K = k) = \frac{\lambda^k e^{-\lambda}}{k!}. \tag{4.18}$$

The burst durations are drawn from a uniform distribution,

$$D_j \sim \text{Uniform}(D_{\min}, D_{\max}), \tag{4.19}$$

and the burst magnitudes are drawn from a log-normal distribution,

$$M_j \sim \text{LogNormal}(\mu_{\log,A}, \sigma_{\log,A}). \tag{4.20}$$

In the discrete implementation, these bursts may be shorter than the 30-second telemetry window. Therefore, each burst contribution is assigned to telemetry windows using fractional overlap between the burst interval $[t_j, t_j + D_j)$ and the affected telemetry windows.

4.2 Replay Telemetry

In order to simulate baseline and compromised network traffic, in some instances, the nodes were configured to replay the baseline telemetry alone as well as with integrated threat signatures. In this manner, threat detection modeling was validated in a more realistic environment, in which the baseline traffic and threat signatures were directly injected into real-time network traffic and analyzed from the resulting telemetry.

Replay telemetry was performed using a six-rank MPI configuration across the NSM lab nodes. The distributed environment consisted of one orchestration node, four sensor node, and one injector node. The orchestration node managed the MPI workflow including reading the baseline traffic models constructed in sec. (4.1), coordinating sensor nodes and collecting and outputting telemetry measurements. The sensor nodes represented the hypothetical R&D workers and replayed the sensor specific Tx/Rx and measured interface byte counters. The injector node, was the traffic endpoint/source for the controlled Tx/Rx replay.

Packet generation was performed using persistent TCP streams created with nc. For transmitted replay, each worker opened a TCP stream to the injector on a configured TX replay port. For received replay, the injector opened TCP streams back to the workers on configured RX replay ports. Linux traffic control was then used to rate-shape the TCP streams on a per-window basis, so packet generation and rate control were separated: nc supplied the traffic stream, while tc imposed the target replay rate. Additionally, because tc was applied only to the replay TCP streams, unrelated SSH, MPI control traffic, and NFS traffic were not restricted by the shaped classes. This was particularly important in the present NSM lab because the runtime files and input data for each node were stored on a common share exported by the storage node.

4.3 Mathematical Threat Detection Modeling

The detection architecture treats network telemetry as a multidimensional, time-varying real-valued signal containing both environmental noise and potential threat signatures. To distinguish routine operational traffic from structured exfiltration tactics, the mathematical processing engine analyzes telemetry streams across multiple layers, ranging from localized point-in-time statistical deviations on individual hosts to distributed, network-wide spatial, temporal, and multiscale correlations. The telemetry observables themselves are real-valued throughout; however, some analytic transformations may introduce intermediate mathemati-

cal representations, such as complex-valued wavelet coefficients, before producing real-valued detection metrics.[4]

4.3.1 Localized Attack-Likelihood Detection Model

The normalized, system-wide attack-likelihood is calculated from independent localized attack-likelihood estimates. It is defined as:

$$P_{\text{sys}}(t) = 1 - \prod_{i=1}^{N} \left[1 - P_i(t)\right], \tag{4.21}$$

where N is the number of sensors and $P_i(t)$ is the local attack-likelihood for sensor i at time t. This expression corresponds to the likelihood that at least one local detector identifies attack-like behavior, under the simplifying assumption that the localized detections are independent. The local attack-likelihood is defined as the expectation-style weighted combination

$$P_i(t) = \sum_{k=1}^{4} w_k x_i^{(k)}(t), \tag{4.22}$$

where $x_i^{(k)}(t)$ is the k-th anomaly component for sensor i, and the weights are constrained to satisfy

$$w_k \geq 0, \qquad \sum_{k=1}^{4} w_k = 1. \tag{4.23}$$

Accordingly, the local attack-likelihood is bounded on the interval

$$0 \leq P_i(t) \leq 1. \tag{4.24}$$

Here, $x_i^{(k)}(t) \in [0, 1]$ is the k-th normalized anomaly component for sensor i. In the present study, $k \in \{1, 2, 3, 4\}$ anomaly components are used to define the system-wide attack likelihood: 1) time, 2) magnitude, 3) persistence, and 4) directionality.

For each component, we choose a hyperbolic tangent mapping for the normalized anomaly component. While $\tanh : \mathbb{R} \to (-1, 1)$ rather than $[0, 1]$, it has several features that make it well suited to the present study. For example, the baseline maps naturally to 0 without

requiring an additional shift parameter. Moreover, the response saturates smoothly for large arguments, so increasingly large differences from baseline produce progressively smaller changes in the mapped anomaly value. Furthermore, as seen below, only the positive branch of the mapping is incorporated into the anomaly calculation, as the negative branch is, in general, not relevant to identifying attack signatures.

The time anomaly component measures whether the total traffic observed at time t is unusually large relative to the expected baseline activity for that sensor. The time component is modeled as

$$x_i^{(1)}(t) = \max\Big(0, \tanh\Big(k_t z_i^{(t)}(t)\Big)\Big), \tag{4.25}$$

where the normalized excess of observed activity above the expected baseline is defined as

$$z_i^{(t)}(t) = \frac{A_i(t) - B_i^{(A)}(t)}{B_i^{(A)}(t)}, \tag{4.26}$$

and the total observed activity for sensor i is defined as

$$A_i(t) = S_i(t) + R_i(t). \tag{4.27}$$

In Eqs. (4.25)–(4.27), k_t is the timing-sensitivity parameter; $S_i(t)$ and $R_i(t)$ are the transmitted and received traffic, respectively; and $B_i^{(A)}(t)$ is the expected total-activity baseline for sensor i.

The magnitude anomaly component measures whether the transmitted or received traffic observed at time t is unusually large relative to the expected directional baseline activity for that sensor. The magnitude component is modeled as

$$x_i^{(2)}(t) = \max\Big(x_i^{(S)}(t), x_i^{(R)}(t)\Big), \tag{4.28}$$

where

$$x_i^{(S)}(t) = \max\Big(0, \tanh\Big(k_S z_i^{(S)}(t)\Big)\Big), \tag{4.29a}$$

$$x_i^{(R)}(t) = \max\Big(0, \tanh\Big(k_R z_i^{(R)}(t)\Big)\Big), \tag{4.29b}$$

and the normalized excesses are given by

$$z_i^{(S)}(t) = \frac{S_i(t) - B_i^{(S)}(t)}{B_i^{(S)}(t)}, \tag{4.30a}$$

$$z_i^{(R)}(t) = \frac{R_i(t) - B_i^{(R)}(t)}{B_i^{(R)}(t)}. \tag{4.30b}$$

In Eqs. (4.28)–(4.30b), analogous notation is used as in Eqs. (4.25)–(4.27). Notably, the magnitude component, Eq. (4.28), is evaluated as the maximum between the transmission and reception anomaly sub-components. This choice obviates the need for calibration of any mixing parameter and ensures that a strong anomaly in one direction is not obfuscated by nominal quantities in the other direction.

The persistence component measures whether elevated activity continues across a larger number of consecutive anomalous intervals. It captures sustained or staged behavior that may not appear suspicious at a single time, but is identified as anomalous by consecutive repetition. The persistence component is given by

$$x_i^{(3)}(t) = \max\Big(0, \tanh\Big(k_p z_i^{(p)}(t)\Big)\Big) \tag{4.31}$$

and the normalized excess is given as

$$z_i^{(p)}(t) = \frac{q_i(t - t_0) - T}{T}. \tag{4.32}$$

In Eq. (4.32), $q_i(t - t_0)$ is the time interval of continuous anomalies beginning at trigger time t_0 and continuing through time t, and T is the persistence threshold. In the present study, t_0 is determined from the time and magnitude anomaly components, so $q_i(t - t_0)$ is not independently selected.[5]

The directionality anomaly component measures whether the balance between transmitted

and received traffic at time t is inconsistent with the expected balance for that sensor. The directionality component is given as

$$x_i^{(4)}(t) = \begin{cases} \tanh\left(k_d z_i^{(d)}(t)\right), & S_i(t) + R_i(t) > 0, \\ 0, & S_i(t) + R_i(t) = 0. \end{cases} \tag{4.33}$$

where the absolute directionality mismatch is given by

$$z_i^{(d)}(t) = \left| D_i(t) - D_i^*(t) \right|, \tag{4.34}$$

and

$$D_i(t) = \frac{S_i(t) - R_i(t)}{S_i(t) + R_i(t)}. \tag{4.35}$$

Here, $D_i(t)$ is the observed signed directionality and $D_i^*(t)$ is the expected signed directionality. It should be noted that a consistent approach was taken here as with the other components above. Namely, $x_i^{(4)}(t) = \max(0, \tanh(k_d z_i^{(d)}(t))) = \tanh(k_d z_i^{(d)}(t))$, because $z_i^{(d)}(t) = |D_i(t) - D_i^*(t)| \geq 0$. Additionally, the piecewise approach is introduced into Eq. (4.33) to account for the resulting singularity where there is no activity.

4.3.2 Lagged Threat-Signature Correlation Detection

While the local likelihood model above evaluates whether a given sensor is anomalous relative to its own baseline, here we address distributed attacks, which generally exhibit coordinated temporal structure across multiple sensors. In particular, we attempt to detect distributed attacks through lagged, or time-translated, correlations between sensors i and j, which are defined as

$$C_{ij}(\tau) = \frac{1}{|\Omega_\tau|} \int_{\Omega_\tau} Z_i(t)\, Z_j(t+\tau)\, dt. \tag{4.36}$$

In the above equation, $Z_i(t)$ is the normalized threat-signature signal for sensor i, τ is the continuous time lag, and $\Omega_\tau \subseteq \Omega$ is the portion of the analysis interval for which both t and $t + \tau$ lie inside the analysis window $\Omega = [t_a, t_b]$. The length of the analysis window is

$|\Omega| = t_b - t_a$. With this convention, positive τ means that the threat-signature structure on sensor j follows the threat-signature structure on sensor i by time τ.

The normalized threat-signature signal is given by

$$Z_i(t) = \frac{s_i(t) - \mu_i}{\sigma_i}, \tag{4.37}$$

where

$$\mu_i = \frac{1}{|\Omega|} \int_\Omega s_i(t)\, dt, \tag{4.38a}$$

$$\sigma_i = \left[\frac{1}{|\Omega|} \int_\Omega \left(s_i(t) - \mu_i\right)^2 \, dt \right]^{1/2}. \tag{4.38b}$$

where μ_i and σ_i are the mean and standard deviation of the threat-signature signal over the analysis window. In Eq. (4.37), $s_i(t)$ is the threat-signature signal for sensor i, which identifies telemetry departing from baseline.[6] The transmitted and received threat-signature components are defined as

$$\Delta S_i(t) = S_i(t) - B_i^{(S)}(t), \tag{4.39a}$$

$$\Delta R_i(t) = R_i(t) - B_i^{(R)}(t). \tag{4.39b}$$

Here, $S_i(t)$ and $R_i(t)$ refer to transmitted and received traffic for sensor i, respectively, and $B_i^{(S)}(t)$ and $B_i^{(R)}(t)$ refer to the corresponding baseline values. The signal $s_i(t)$ may be selected from the transmitted or received threat-signature component, depending on the directionality of the attack signature under analysis.

Referring to Eq. (4.36), the peak lag between sensors i and j is the value of $\tau \in [-\tau_{\max}, \tau_{\max}]$ for which the lagged correlation is maximized,

$$C_{ij}(\tau_{ij}^*) = \max_{\tau \in [-\tau_{\max}, \tau_{\max}]} C_{ij}(\tau), \tag{4.40}$$

with corresponding peak correlation

$$C_{ij}^{*} = C_{ij}\left(\tau_{ij}^{*}\right). \tag{4.41}$$

A pair of sensors is considered strongly correlated when

$$C_{ij}^{*} \geq C_{\mathrm{th}}, \tag{4.42}$$

where C_{th} is the correlation threshold. These strong pairwise correlations define a timing graph whose vertices are sensors and whose edges are lagged threat-signature correlations. Equivalently, the recovered peak lags may be represented as a lag matrix.

For a coordinated distributed signature, the pairwise lags should be mutually consistent. For an ideal continuous signature for a triplet of sensors (i, j, ℓ), exact lag consistency requires

$$\tau_{ij}^{*} + \tau_{j\ell}^{*} = \tau_{i\ell}^{*}. \tag{4.43}$$

In measured telemetry, the recovered peak lags need not satisfy Eq. (4.43) exactly because of finite sampling, measurement noise, replay artifacts, and the finite width of the threat signatures. We therefore test lag consistency using a tolerance,

$$\left|\tau_{ij}^{*} + \tau_{j\ell}^{*} - \tau_{i\ell}^{*}\right| \leq \epsilon_{\tau}, \tag{4.44}$$

where ϵ_{τ} is the allowed lag-consistency tolerance.

The coordination fraction is then defined as

$$F_{\mathrm{coord}} = \frac{N_{\mathrm{consistent}}}{N_{\mathrm{checked}}}, \tag{4.45}$$

where N_{checked} is the number of strongly connected sensor triplets and $N_{\mathrm{consistent}}$ is the number of those triplets satisfying Eq. (4.44).

Finally, the many-body coordination alarm is defined as

$$\mathcal{A}_{\mathrm{coord}} = \begin{cases} 1, & N_{\mathrm{edges}} \geq N_{\mathrm{edges}}^{\mathrm{min}}, \quad N_{\mathrm{checked}} > 0, \quad F_{\mathrm{coord}} \geq F_{\mathrm{coord}}^{\mathrm{min}}, \\ 0, & \text{otherwise.} \end{cases} \tag{4.46}$$

Here, N_{edges} is the number of strong unordered sensor pairs, $N_{\text{edges}}^{\min}$ is the minimum number of required strong edges, and $F_{\text{coord}}^{\min}$ is the minimum required fraction of lag-consistent triplets. This detector therefore identifies coordinated timing structure across sensors, rather than relying only on the magnitude of the local anomaly likelihoods $P_i(t)$.

4.3.3 Wavelet-Coherence Analysis

To examine coordinated multiscale structure in weak or geometrically discordant threat signatures, a wavelet-coherence analysis was applied to the inferred threat-signature components. For sensor s, the TX and RX threat-signature components were defined as

$$\Delta x_s^{\text{TX}}(t) = x_{s,\text{obs}}^{\text{TX}}(t) - x_{s,\text{base}}^{\text{TX}}(t), \tag{4.47}$$

$$\Delta x_s^{\text{RX}}(t) = x_{s,\text{obs}}^{\text{RX}}(t) - x_{s,\text{base}}^{\text{RX}}(t). \tag{4.48}$$

Because the threat signatures may contain both TX and RX components, the scalar threat-signature signal used for wavelet analysis was taken to be the bidirectional threat-signature magnitude,

$$q_s(t) = \sqrt{\left[\Delta x_s^{\text{TX}}(t)\right]^2 + \left[\Delta x_s^{\text{RX}}(t)\right]^2}. \tag{4.49}$$

This scalarization intentionally discards TX/RX directionality in favor of detecting shared multiscale energy across sensors. Directional asymmetry is evaluated separately by the localized attack-likelihood model.

Let $\psi(t)$ denote the mother wavelet. The scaled and translated analyzing wavelet is

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}}\,\psi\left(\frac{t-b}{a}\right), \tag{4.50}$$

where $a > 0$ is the wavelet scale and $b \in \mathbb{R}$ is the time translation. In the present implementation, $\psi(t)$ is a complex Morlet wavelet, so the telemetry signals $q_s(t)$ are real-valued while the wavelet coefficients are complex-valued intermediate quantities.

The continuous wavelet transform of $q_s(t)$ is then

$$W_s(a,b) = \int_{-\infty}^{\infty} q_s(t)\,\psi_{a,b}^*(t)\,dt, \tag{4.51}$$

or equivalently,

$$W_s(a,b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} q_s(t)\,\psi^*\left(\frac{t-b}{a}\right)\,dt. \tag{4.52}$$

Here, the superscript * denotes complex conjugation.

For two sensors i and j, the cross-wavelet transform is defined as

$$W_{ij}(a,b) = W_i(a,b)\, W_j^*(a,b). \tag{4.53}$$

The squared wavelet coherence between sensors i and j is then

$$R_{ij}^2(a,b) = \frac{\left|S(a^{-1}W_{ij}(a,b))\right|^2}{\left|S(a^{-1}|W_i(a,b)|^2)\right|\,\left|S(a^{-1}|W_j(a,b)|^2)\right|}, \tag{4.54}$$

where $S(\cdot)$ denotes smoothing in time and scale. The coherence satisfies

$$0 \le R_{ij}^2(a,b) \le 1, \tag{4.55}$$

with larger values indicating stronger localized agreement in the wavelet domain.

To extend the pairwise coherence to a distributed, many-sensor measure, the coherence values were averaged over all unordered sensor pairs,

$$M_{\text{all}}(a,b) = \frac{2}{N(N-1)} \sum_{i<j} R_{ij}^2(a,b), \tag{4.56}$$

where N is the number of sensors. For $N = 4$, this becomes

$$M_{\text{all}}(a,b) = \frac{1}{6}\left(R_{12}^2 + R_{13}^2 + R_{14}^2 + R_{23}^2 + R_{24}^2 + R_{34}^2\right). \tag{4.57}$$

The two-dimensional coherence field $M_{\text{all}}(a,b)$ provides a time-scale representation of distributed coordination. To obtain a one-dimensional detection curve, the coherence field was averaged over a selected scale band $\mathcal{A}$,

$$C_{\text{all}}(b) = \frac{1}{|\mathcal{A}|} \int_{\mathcal{A}} M_{\text{all}}(a,b)\, da, \tag{4.58}$$

where $|\mathcal{A}|$ is the measure of the scale interval. This produces a time-dependent multiscale coherence score.

A wavelet-coherence alarm was defined using both amplitude and persistence. Let θ denote the coherence threshold. The coherence support set is

$$\Omega_\theta = \{b : C_{\text{all}}(b) > \theta\}. \tag{4.59}$$

The area above threshold is

$$A_\theta = \int \max[C_{\mathrm{all}}(b) - \theta,\, 0]\; db. \tag{4.60}$$

The longest contiguous duration above threshold is

$$D_\theta = \sup_{\mathcal{I} \subseteq \Omega_\theta} |\mathcal{I}|, \tag{4.61}$$

where $\mathcal{I}$ denotes a connected time interval contained in Ω_θ, and $|\mathcal{I}|$ is its duration.

The wavelet-coherence alarm condition was therefore

$$A_\theta \geq A_{\mathrm{min}} \quad \text{and} \quad D_\theta \geq D_{\mathrm{min}}, \tag{4.62}$$

where A_{min} is the minimum required coherence-area support and D_{min} is the minimum required duration above threshold.

5. Results and Discussion

The proposed distributed threat-detection architecture using both replayed network telemetry and synthetic threat-signature experiments is evaluated. The replay telemetry established the behavior of the detection framework under controlled home-lab traffic conditions, while the synthetic analyses provided a broader evaluations of localized anomaly likelihoods, lagged coordination structure, and multiscale wavelet coherence. Together, these results assessed whether the proposed models could distinguish routine network traffic from structured exfiltration-like behavior across multiple sensors.

5.1 Replay Telemetry

Synthetic baselines were constructed as described in Sec. 4.1, both with and without injected threat signatures, and the replayed telemetry was validated against the corresponding target signals. The parameters used to calculate the synthetic baselines are shown in Appendix A.[7] The results of the baseline construction are shown in Figs. 5.1 and 5.2.

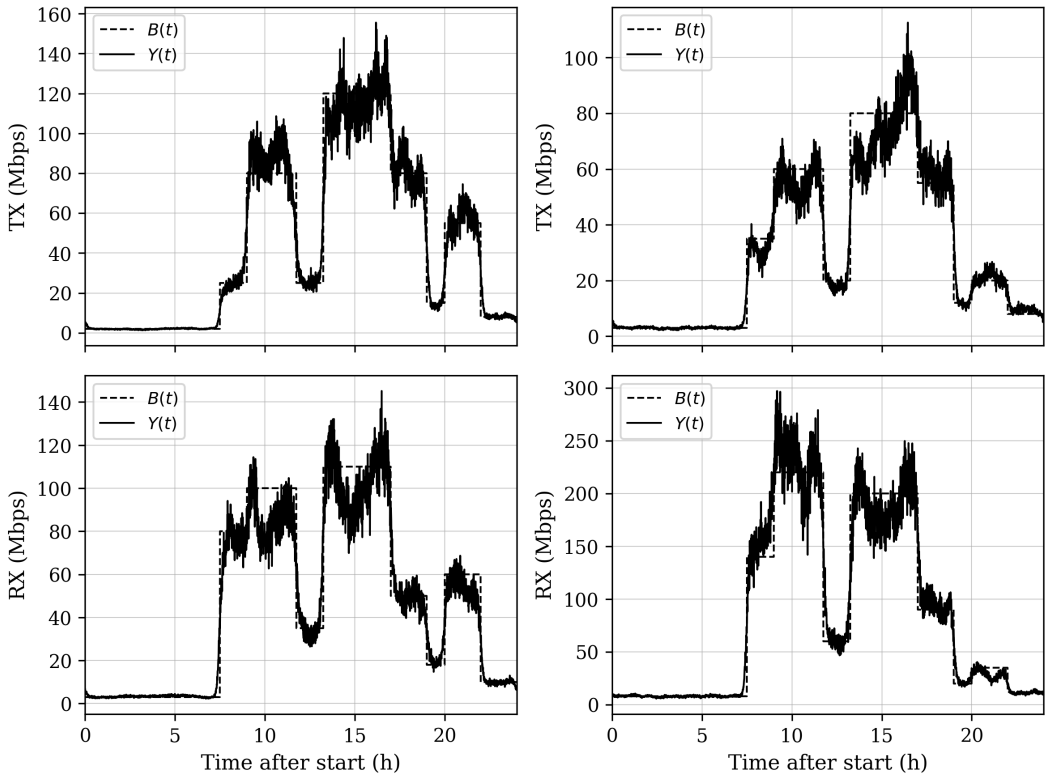


Figure 5.1: Baseline traffic construction. Top: nominal and smoothed transmitted-byte baselines for sensors 1 (left) and 2 (right). Bottom: nominal and smoothed received-byte baselines for sensors 1 (left) and 2 (right).

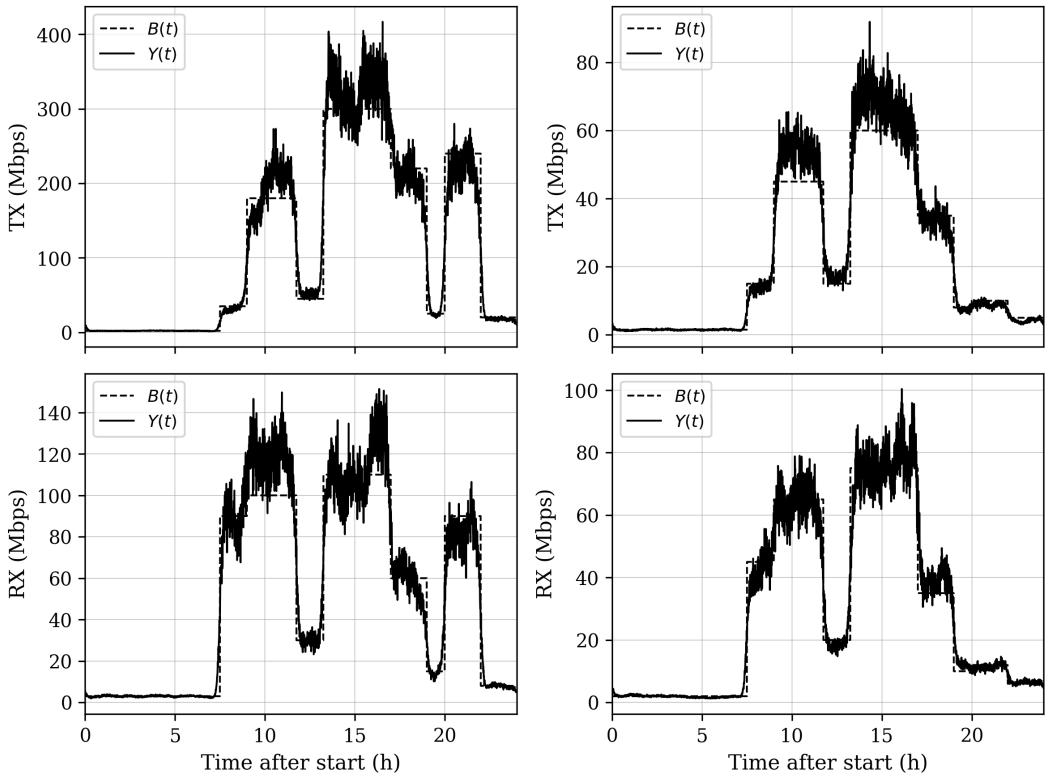


Figure 5.2: Baseline traffic construction. Top: nominal and smoothed transmitted-byte baselines for sensors 3 (left) and 4 (right). Bottom: nominal and smoothed received-byte baselines for sensors 3 (left) and 4 (right).

The figures demonstrate that the construction preserves the intended class-dependent daily structure while replacing the artificial plateaus of the nominal baseline with nontrivial stochastic variation. The resulting signals retain the expected sensor-specific transmit and receive behavior, while providing a more realistic baseline for replay and later anomaly-detection experiments.

Following the synthetic baseline construction, the telemetry replay was validated. The parameters used for the telemetry replay are shown in Appendix B. The telemetry replay was taken over a two-hour window of the 24-hour day, namely from approximately 10:00 AM to 12:00 PM. From Figs. 5.1 and 5.2, this interval corresponds to an appreciably busy period of workday network activity. The results of the baseline-only telemetry replay are shown in Table 5.1.

Sensor	**Direction**	**Mean Ratio**	**Std. Dev.**	**Max. Deviation**
1	TX	0.963	0.001	3.82%
1	RX	1.000	0.004	1.16%
2	TX	0.971	0.002	3.19%
2	RX	0.997	0.004	0.93%
3	TX	0.960	0.000	4.05%
3	RX	1.006	0.004	1.19%
4	TX	0.966	0.001	3.57%
4	RX	0.998	0.004	0.83%

Table 5.1: Replay fidelity for the high-throughput chaotic baseline segment. Ratios are measured traffic divided by target traffic. Maximum deviation is computed as the largest absolute deviation from unity. The TX replay exhibits a systematic underperformance of approximately 3–4%, while RX replay remains close to unity for all sensors.

Table 5.1 shows that the measured replay traffic closely tracked the target baseline after the initial startup transient was removed. The TX mean ratios remained within approximately 4.0% of unity for all sensors, while the RX mean ratios remained within approximately 0.6% of unity. The maximum observed deviations were below approximately 4.1%, indicating good replay fidelity over the analyzed interval. The remaining deviations are consistent with expected real-system variability from TCP behavior, Linux scheduling, traffic shaping, and shared network resources. A representative example of the replay validation is shown in Fig. 5.3.

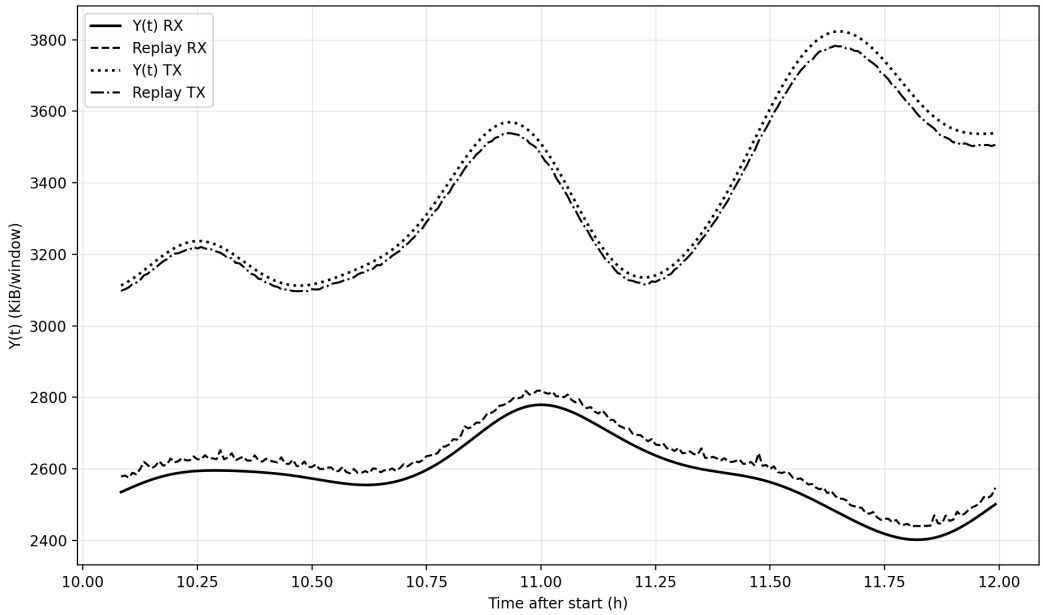


Figure 5.3: Baseline replay performance for representative sensor 3.

As can be seen from the figure, the replay results showed a small but systematic directional bias: measured RX replay exceeded the target curves, while measured TX replay underperformed the target curves. This bias is consistent with real-system effects introduced by TCP stream generation, Linux traffic shaping, interface-counter measurement, ACK/control traffic, and nonidentical sender-side shaping paths for the TX and RX replay directions. Notably, changing the HTB rate-to-quantum parameter, not shown here, did not materially alter the bias. Nevertheless, for the purposes of the present study, the observed replay fidelity is sufficient to validate the replay framework as a realistic input layer for subsequent mathematical threat-model evaluation. Although further engineering refinements, such as replacing `nc` with a more precise traffic generator, could likely reduce the remaining directional replay bias, such refinements are not expected to materially affect the qualitative detection behavior examined here.

A second replay experiment was then performed using synthetic baseline telemetry with injected threat signatures. The threat-signature characteristics are displayed in Table 5.2.

Threat	Sensor	Threat Interval	Wall-Clock Interval	TX Rate	TX Volume
1	1	1801–1809	15:00:00–15:04:30	15 Mbps	450 MB
2	2	1805–1813	15:02:00–15:06:30	15 Mbps	450 MB
3	3	1809–1817	15:04:00–15:08:30	15 Mbps	450 MB
4	4	1813–1821	15:06:00–15:10:30	15 Mbps	450 MB
				Total TX exfiltration volume	**1.8 GB**

Table 5.2: Threat-signature configuration for the compressed distributed exfiltration replay. The telemetry window length is 30 seconds. Each threat is TX-only and is staggered by two minutes relative to the preceding sensor.

Table 5.2 shows that the threat signature was modeled as a distributed exfiltration event across all sensors. Each sensor-specific signature was modeled as a plateau with a small overlap between the previous and subsequent signatures. The plateau magnitude was intentionally kept modest so that the attack would not dominate the baseline traffic by simple volume alone. For sensors 1 to 4, the 15 Mbps plateau corresponded to approximately 12.7%, 17.5%, 4.70%, and 19.4%, respectively, of the maximum baseline TX rate over the same interval. The relatively small 1.8 GB total exfiltration size is intended to correspond to a small database.

The results of the baseline-plus-threat telemetry replay are shown in Table 5.3.

Sensor	Direction	Mean Ratio	Std. Dev.	Max. Deviation
1	TX	0.963	0.001	3.87%
1	RX	1.000	0.005	1.70%
2	TX	0.971	0.001	3.18%
2	RX	0.997	0.006	2.10%
3	TX	0.960	0.000	4.05%
3	RX	1.006	0.006	1.77%
4	TX	0.965	0.001	3.76%
4	RX	0.998	0.004	1.02%

Table 5.3: Replay fidelity for the compressed baseline-plus-threat replay segment. Ratios are measured traffic divided by target traffic, where the target is constructed as the baseline plus the injected threat signature. Maximum deviation is computed as the largest absolute deviation from unity.

The baseline-plus-threat replay showed fidelity comparable to the baseline-only replay in Table 5.1. After removal of the initial startup transient, the measured telemetry closely

tracked the synthesized target signal, including the injected threat intervals. The TX mean ratios again remained within approximately 4.0% of unity for all sensors, while the RX mean ratios remained close to unity. The maximum observed deviations remained below approximately 4.1%, indicating that the injected threat signatures did not materially degrade replay fidelity. A representative example of the baseline-plus-threat replay validation is shown in Fig. 5.4.

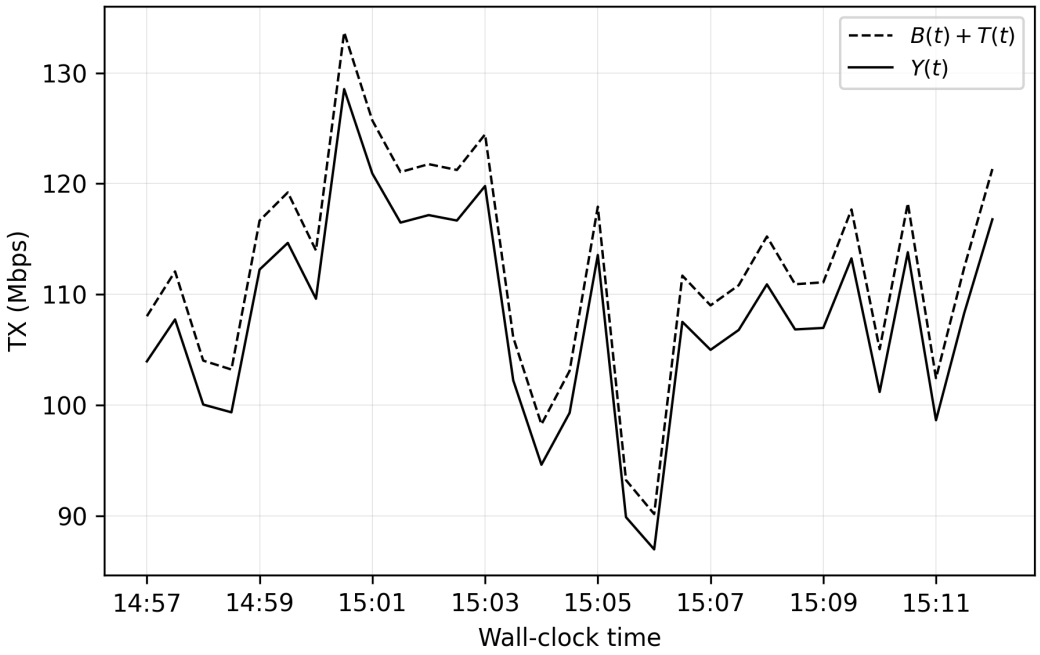


Figure 5.4: Baseline-plus-threat replay performance for representative sensor 1.

As was the case with the baseline-only telemetry, the baseline-plus-threat replay showed a small but systematic directional bias: measured RX replay exceeded the target curves, while measured TX replay underperformed the target curves. For the purposes of the present study, however, the observed replay fidelity is sufficient to validate the replay framework as a realistic input layer for subsequent mathematical threat-model evaluation.

5.2 Mathematical Threat Detection Analysis

Having validated that the replay framework can reproduce both baseline-only and baseline-plus-threat telemetry with acceptable fidelity, the next step was to evaluate the mathematical detection models' ability to identify the injected threat signatures. In particular, the performance of the localized attack-likelihood, lagged threat-signature correlation, and wavelet-

coherence detection layers were assessed. The goal was not merely to determine whether the injected traffic increased total volume, but whether the proposed detection layers recovered the intended structure of the distributed exfiltration signature across sensors, time offsets, and multiscale behavior.

5.2.1 Synthetic Telemetry Detection Analysis: Lagged Correlation

To further evaluate the threat-detection models, additional analyses were conducted using synthetic telemetry generated directly from the baseline and threat-signature models. Synthetic telemetry enabled a broader exploration of attack signatures while avoiding the wall-clock timing constraints imposed by physical replay in the home-lab environment. This allowed the detection models to be tested over a wider range of threat durations, amplitudes, and temporal structures than would be practical using replay experiments alone.

To further test the threat-detection model, a second distributed exfiltration scenario was constructed using plateau-shaped TX threat signatures. As in the previous replay experiment, the use of plateau functions provides a controlled threat geometry with well-defined start times, end times, magnitudes, and total exfiltration volumes. However, in this case the threat signatures were designed to test a more challenging distributed timing structure. Rather than imposing a uniform delay between adjacent sensors, the threat activations were displaced according to a Fibonacci-lag pattern. The adjacent sensor delays were 8, 5, and 13 telemetry windows, corresponding to 4.0, 2.5, and 6.5 minutes, respectively.

In addition to the nonuniform timing offsets, the threat magnitudes were also varied across sensors. This produces a less idealized distributed signature than the equal-amplitude replay case: the detector must identify a coordinated timing structure even though the local TX contributions are not identical in magnitude. The resulting event is therefore useful for testing whether the lagged threat-signature correlation model is sensitive to the shape and timing of distributed threat signatures, rather than simply responding to simultaneous or equal-magnitude anomalies.

A summary of the Fibonacci-lag threat signatures is displayed in Table 5.4. Complete results can be found in Appendix **??**.

Threat	Sensor	Threat Interval	Wall-Clock Interval	TX Rate	TX Volume
1	1	1300–1309	10:49:30–10:54:30	4.48 Mbps	160.15 MiB
2	2	1308–1317	10:53:30–10:58:30	2.61 Mbps	93.39 MiB
3	3	1313–1322	10:56:00–11:01:00	11.29 Mbps	403.58 MiB
4	4	1326–1335	11:02:30–11:07:30	2.60 Mbps	92.87 MiB
Total TX exfiltration volume					**750.00 MiB**

Table 5.4: Summary of the Fibonacci-lag distributed TX exfiltration signatures.

Referring to the table, the total injected TX volume is 750.00 MiB, distributed unevenly across the four sensors. It is also notable that the TX rates are substantially smaller than those used in the earlier replay telemetry threat signatures. As a result, this scenario provides a useful test of the model's ability to detect structured coordination in a lower-rate, nonuniform, distributed exfiltration pattern.

As was the case with the threat telemetry replay, the localized likelihood model assigned only a LOW likelihood tier to the Fibonacci-lag exfiltration scenario. A summary of the results is displayed in Table 5.5. Complete results are provided in Appendix D.

Quantity	**Result**
System-wide peak likelihood	LOW, $P_{\text{sys}} = 0.4185$
System-wide detection interval	1306–1335; 10:52:30–11:07:30
Localized sensor likelihoods	LOW for all sensors
Localized peak likelihood range	0.2937–0.3082
Minimum off-diagonal support-normalized correlation	1.0000
Maximum off-diagonal support-normalized correlation	1.0000
Fixed-N correlation range	0.9965
Strong unordered sensor pairs	6
Recovered adjacent-sensor delays	$8, 5, 13$ windows
Recovered adjacent-sensor delays	4.0, 2.5, 6.5 min
Consistent timing triplets	4
Coordination fraction F_{coord}	1.000
Many-body coordination alarm	Alarm

Table 5.5: Summary of the localized likelihood and lagged threat-signature correlation analysis for the Fibonacci-lag baseline-plus-threat replay telemetry.

Referring to Table 5.5, the system-wide peak probability was $P_{\text{sys}} = 0.4185$, occurring at replay window 1317, corresponding to 10:58:00. Similarly, each individual sensor remained

in the LOW likelihood tier, with peak local probabilities of 0.3018, 0.2937, 0.3082, and 0.2998 for sensors 1–4, respectively. The detected local intervals were shorter than the full injected threat intervals, indicating that the local model responded only to the strongest portions of each plateau. Thus, when considered only as isolated per-sensor anomalies, the Fibonacci-lag signatures were not classified as strong attacks. This is consistent with the lower TX rates and unequal threat magnitudes used in this test case, and it motivates the need for the lagged threat-signature correlation analysis to identify the distributed coordination structure.

The lagged threat-signature correlation detector performed strongly on the Fibonacci-lag scenario. Although the localized likelihood scores remained in the LOW tier, the support-normalized lagged correlations were essentially unity for all sensor pairs, indicating that the TX threat signatures had nearly identical plateau structure after lag compensation. The recovered lag matrix correctly identified the imposed Fibonacci timing pattern, with adjacent-sensor delays of 8, 5, and 13 windows, corresponding to 4.0, 2.5, and 6.5 minutes, respectively. The non-adjacent delays were also recovered consistently, with pairwise offsets of 13, 18, and 26 windows. All six unordered sensor pairs exceeded the correlation threshold, all four checked timing triplets satisfied the pairwise-lag consistency condition, and the coordination fraction was $F_{\text{coord}} = 1.000$. Consequently, the many-body coordination model raised an alarm, demonstrating that the distributed timing structure was detectable even when the local attack-likelihood model classified the event as weak.

To further test the threat-detection model, a mixed-signal exfiltration scenario was analyzed. This scenario was designed to reduce the effectiveness of the localized likelihood model while preserving a coordinated distributed structure. First, both TX and RX components were included so that the TX/RX ratio was partially balanced, thereby reducing the directionality skew associated with a purely outbound exfiltration event. Second, the peak TX magnitudes were chosen to be small relative to the local baseline traffic, ranging from approximately 2% to 5% of the corresponding baseline TX rate. Finally, geometric discordance was introduced by assigning a different temporal signature to each sensor, including a parabolic pulse, an exponential ramp, a sinusoidal swell, and a stochastic high-variance burst. These mismatched and partially overlapping signatures were intended to stress-test the lagged threat-signature correlation model by reducing simple shape agreement while preserving an underlying coordinated timing structure. The resulting threat signatures are shown in Fig. 5.5, and their key properties are summarized in Table 5.6.

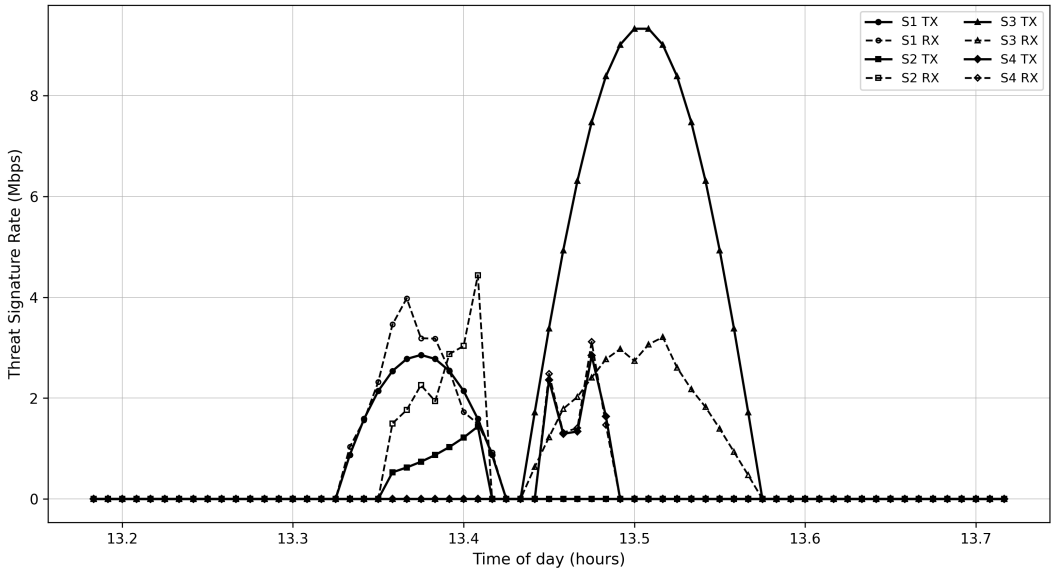


Figure 5.5: Mixed-signal threat signatures. Solid curves denote TX components and dashed curves denote RX components. The four sensors exhibit intentionally mismatched signal geometries: parabolic pulse, exponential ramp, sinusoidal swell, and stochastic high-variance burst.

Threat	**Sensor**	**Shape**	**Threat Interval**	**Peak TX Rate**	**Volume**
1	1	Pulse-like	1601–1611; 13:20:00–13:25:30	2.14 Mbps	TX: 81.16 MiB; RX: 90.88 MiB
2	2	Ramp-like	1604–1610; 13:21:30–13:25:00	0.87 Mbps	TX: 23.06 MiB; RX: 63.69 MiB
3	3	Broad swell	1614–1629; 13:26:30–13:34:30	6.89 Mbps	TX: 361.44 MiB; RX: 115.31 MiB
4	4	Short burst	1615–1619; 13:27:00–13:29:30	1.64 Mbps	TX: 33.89 MiB; RX: 35.04 MiB
Total TX volume					**499.55 MiB**
Total RX volume					**304.91 MiB**
Total bidirectional threat volume					**804.45 MiB**

Table 5.6: Summary of the mixed-signal distributed threat signatures. The reported TX rate is the peak backed-out TX rate over the active support interval for each sensor.

As expected, the threat signatures neutralized the localized threat detection and system-wide detection models. Across all sensors, the non-direction likelihood components ranged from 0.0% to 76.2%. The time and magnitude components remained small, ranging from 4.6%–9.0% and 4.7%–9.5%, respectively, while the persistence component ranged from 0.0%–76.2%. The directionality component was effectively neutralized by the TX/RX masking, ranging only from 0.3%–0.6% across all sensors. A summary of the results is displayed in Table 5.7. Complete results are provided in Appendix E.

Quantity	**Result**
System-wide peak likelihood	NO indication, $P_{\text{sys}} = 0.2379$
System-wide detection interval	None
Localized sensor likelihoods	NO indication for all sensors
Localized peak likelihood range	0.0279–0.2220
Non-direction component range	0.000–0.762
Directionality component range	0.003–0.006
Minimum off-diagonal support-normalized correlation	0.7016
Maximum off-diagonal support-normalized correlation	0.9499
Fixed-N correlation range	0.6994–0.9460
Strong unordered sensor pairs	6
Recovered adjacent-sensor delays	$2, 14, -4$ windows
Recovered adjacent-sensor delays	1.0, 7.0, -2.0 min
Consistent timing triplets	4
Coordination fraction F_{coord}	1.000
Many-body coordination alarm	Alarm

Table 5.7: Summary of the localized likelihood and lagged threat-signature correlation analysis for the mixed-signal distributed exfiltration telemetry.

The lagged threat-signature correlation detector again performed strongly on the mixed-signal exfiltration scenario. In contrast to the Fibonacci-lag plateau case, the localized and system-wide likelihood models registered *NO indication*: the peak system-wide likelihood was only $P_{\text{sys}} = 0.2379$, and no individual sensor produced a localized detection interval. This is consistent with the construction of the mixed-signal threat signatures, which used relatively small peak magnitudes, bidirectional TX/RX components, and mismatched temporal geometries to suppress the magnitude and directionality components of the local detector.

Despite the absence of a localized likelihood alarm, the lagged threat-signature correlation analysis recovered a coherent distributed timing structure. The support-normalized off-diagonal correlations ranged from 0.7016 to 0.9499, so all six unordered sensor pairs exceeded

the correlation threshold after lag compensation. The recovered lag matrix was nonuniform, reflecting the deliberately mismatched timing and shape structure of the injected signatures, but the pairwise lags remained mutually consistent. All four checked timing triplets satisfied the lag-consistency condition, giving a coordination fraction of $F_{\text{coord}} = 1.000$. As a result, the many-body coordination model raised an alarm even though the local and system-wide likelihood models did not.

This result is significant because the mixed-signal scenario was designed specifically to weaken simple localized indicators. The detector was not responding to a large single-node anomaly or to a purely TX-skewed exfiltration signature. Instead, the alarm was driven by coordinated threat-signature timing across sensors. Thus, the result demonstrates that the lagged threat-signature correlation model can identify distributed structure even when the individual sensor signatures are weak, bidirectional, and geometrically dissimilar.

The Fibonacci-lag and mixed-signal experiments demonstrate that lagged threat-signature correlation can recover distributed timing structure even when localized likelihood scores remain weak. However, both tests still preserve enough pairwise temporal consistency for the lagged-correlation model to succeed. The final synthetic experiment therefore considers a more adversarial threat signature designed to disrupt fixed-lag timing consistency and shift the detectable structure into the multiscale wavelet domain.

5.2.2 Synthetic Telemetry Detection Analysis: Wavelet Coherence

As a final test of the threat-modeling performance, a liquid-skeleton threat signature was constructed to subvert the lagged threat-signature correlation detector. In this scenario, the structural delays and signal geometries across sensors were intentionally made fluid, thereby "liquidizing" the timing graph and disrupting the lag-consistency check used by the lagged correlation detector. A brief summary of the threat-signature characteristics is shown in Table 5.8, and the threat signatures are plotted in Fig. 5.6.

Sensor	Threat Signal Characteristic	Threat Timing	Threat Volume
1	Chirped signal envelope with time-varying frequency.	16:15:00–16:34:30; 19.5 min	115.62 MiB
2	Linear dynamic lag, where the offset increases over the pulse.	16:17:00–16:37:00; 20.0 min	223.75 MiB
3	Oscillating/weaving lag envelope with multiple harmonics.	16:21:00–16:39:30; 18.5 min	167.25 MiB
4	High-frequency stochastic random-walk jitter.	16:19:00–16:33:30; 14.5 min	81.83 MiB
Total bidirectional threat volume			**588.46 MiB**

Table 5.8: Wavelet analysis threat-signal characteristics. The signatures were designed to introduce geometric discordance across sensors, including time-varying frequency content, dynamic lag evolution, harmonic lag modulation, and stochastic high-frequency jitter. Threat volumes include both TX and RX threat-signature components.

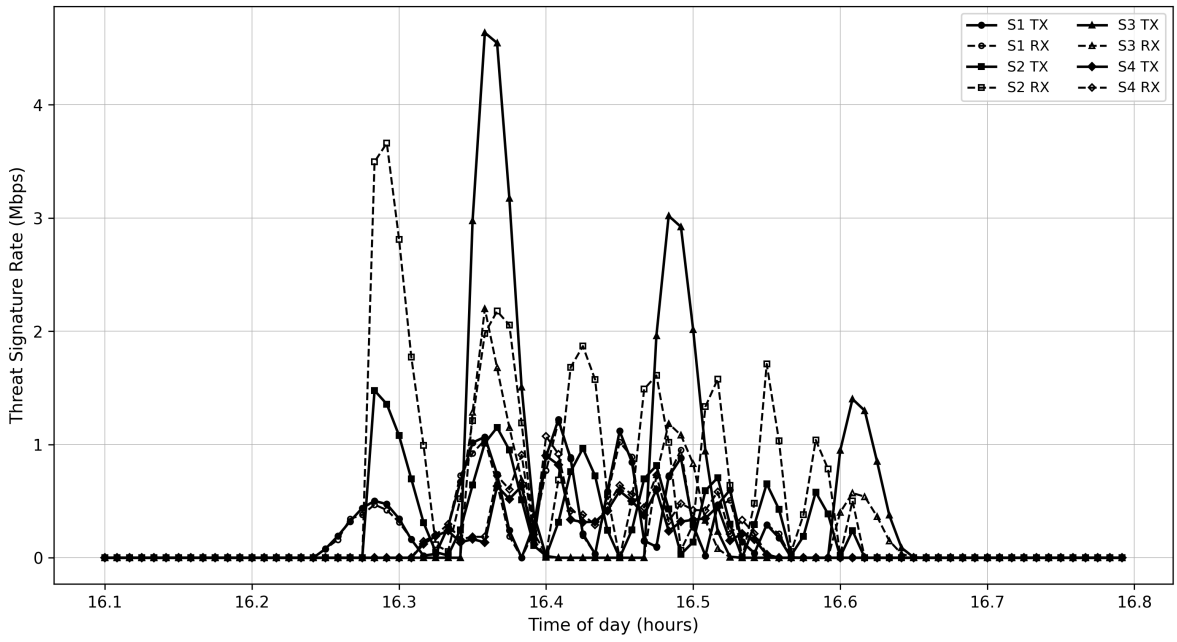


Figure 5.6: Wavelet detection threat signatures. Solid curves denote TX components and dashed curves denote RX components. Marker shape distinguishes sensors 1 to 4. The signatures exhibit weak, irregular, bidirectional structure with overlapping support, including chirped, dynamically lagged, harmonic, and stochastic jitter behavior across the four sensors.

As expected, the localized attack-likelihood and lagged threat-signature correlation detectors were both prevented from raising alarms, as shown in Table 5.9. Consistent with the threat-signature design, the system-wide likelihood model reported *NO indication*, with a peak value of only $P_{\text{sys}} = 0.0458$. Likewise, all four localized sensor detectors remained in the *NO indication* tier, with peak local likelihoods ranging only from 0.0111 to 0.0209. Thus, the event produced essentially no localized magnitude, persistence, or directionality signature strong enough to trigger the likelihood-based detector.

Quantity	**Result**
System-wide peak likelihood	NO indication, $P_{\text{sys}} = 0.0458$
System-wide detection interval	None
Localized sensor likelihoods	NO indication for all sensors
Localized peak likelihood range	0.0111–0.0209
Minimum off-diagonal support-normalized correlation	0.5643
Maximum off-diagonal support-normalized correlation	0.8044
Fixed-N correlation range	0.5609–0.7978
Strong unordered sensor pairs	3
Recovered adjacent-sensor delays	$2, 8, 2$ windows
Recovered adjacent-sensor delays	1.0, 4.0, 1.0 min
Consistent timing triplets	0
Coordination fraction F_{coord}	0.000
Many-body coordination alarm	No-Alarm

Table 5.9: Summary of the localized attack-likelihood and lagged threat-signature correlation results for the liquid-skeleton threat signatures.

The lagged threat-signature correlation model also did not raise a many-body coordination alarm. However, the pairwise correlation results were not negligible: the maximum off-diagonal support-normalized correlation was 0.8044, indicating that at least one sensor pair retained a reasonably strong lag-compensated relationship. The overall correlation structure, however, was too sparse and inconsistent to support a distributed timing alarm, with only three strong unordered edges, zero consistent timing triplets, and a coordination fraction of $F_{\text{coord}} = 0.000$. Therefore, the many-body correlation model correctly remained at *No-Alarm*. This result shows that the threat-signal construction partially preserved pairwise structure while successfully disrupting the consistent many-body timing graph required by the lagged correlation detector.

To determine whether coordinated structure remained detectable in a different representation, the wavelet-coherence model was applied to the same inferred threat signatures. The

scalar signal used for the wavelet analysis was the bidirectional threat magnitude,

$$q_s(t) = \sqrt{[\Delta x_s^{\mathrm{TX}}(t)]^2 + [\Delta x_s^{\mathrm{RX}}(t)]^2},$$

so that both TX and RX components contributed to the multiscale coherence calculation. A complex Morlet wavelet with $\omega_0 = 6$ was used to compute all-pairs wavelet coherence, and the resulting coherence map was collapsed over scales 8–20 to form the detection curve $C_{\mathrm{all}}(t)$. A summary of the wavelet-coherence analysis is provided in Table 5.10.

Quantity	**Result**
Wavelet signal	$\sqrt{TX_{\mathrm{threat}}^2 + RX_{\mathrm{threat}}^2}$, converted to Mbps
Mother wavelet	Complex Morlet, $\omega_0 = 6$
Focus replay window interval	1933–2017
Focus clock interval	16:06:00–16:48:30
CWT scale range	1–32 windows
Collapsed scale band	8–20 windows
Threat volume	TX: 284 MiB; RX: 304 MiB
Total bidirectional threat volume	588 MiB
Coherence threshold θ	0.850
Peak $C_{\mathrm{all}}(t)$	0.940
Peak replay window	2011
Peak clock time	16:45:00
A_θ	0.740 coherence-minutes
D_θ	11.000 min
Minimum required A_θ	0.300 coherence-minutes
Minimum required D_θ	5.000 min
Above-threshold interval 1	1933–1936; 16:06:00–16:08:00
Above-threshold interval 2	1996–2017; 16:37:30–16:48:30
Wavelet-coherence alarm	Alarm

Table 5.10: Summary of the wavelet-coherence analysis for the liquid-skeleton threat telemetry. The analysis used the bidirectional threat magnitude as the scalar signal, computed all-pairs wavelet coherence using a complex Morlet wavelet, and collapsed the coherence map over scales 8–20. The resulting persistence metrics exceeded the configured alarm requirements, indicating coordinated multiscale wavelet structure.

The wavelet-coherence analysis performed well in detecting coordinated structure that was not sufficient to trigger either the localized attack-likelihood model or the lagged threat-signature correlation model. Figure 5.7 shows the scale-collapsed all-pairs coherence curve,

$C_{\mathrm{all}}(t)$. Although the individual threat signatures were weak, bidirectional, and geometrically discordant, the coherence curve exceeded the alarm threshold over sustained intervals. The primary above-threshold interval occurred over replay windows 1996–2017, corresponding to 16:37:30–16:48:30. The peak value was $C_{\mathrm{all}} = 0.940$, occurring at replay window 2011, or 16:45:00. The resulting persistence measures were $A_\theta = 0.740$ coherence-minutes and $D_\theta = 11.0$ minutes, exceeding the configured alarm requirements of $A_\theta \geq 0.300$ coherence-minutes and $D_\theta \geq 5.0$ minutes. Therefore, the wavelet-coherence model raised an *Alarm*.

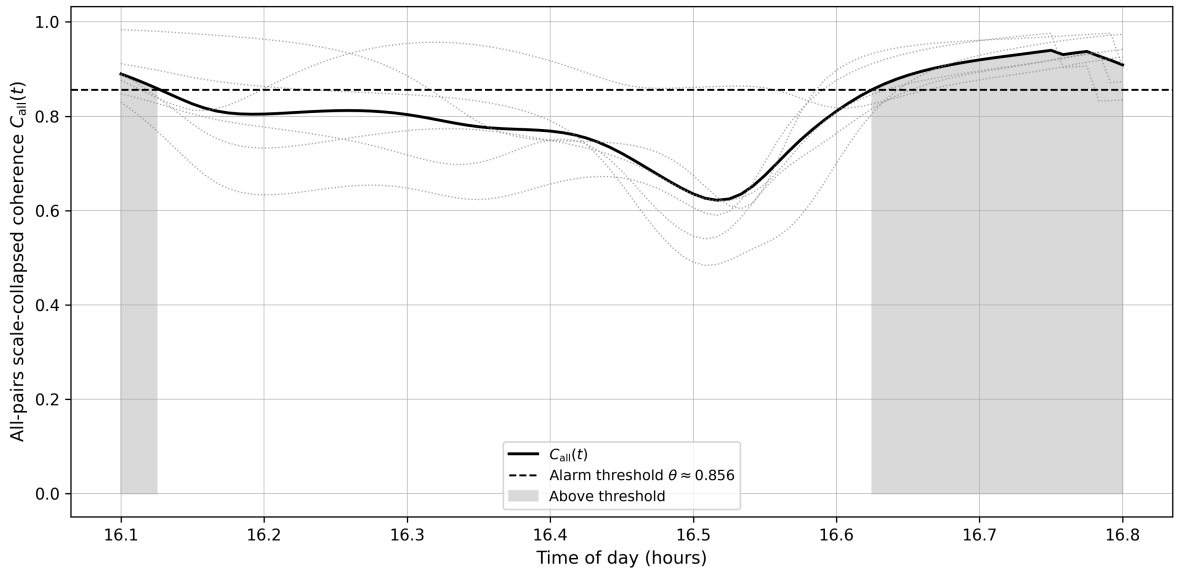


Figure 5.7: Scale-collapsed all-pairs wavelet-coherence detection curve for the liquid-skeleton threat signatures. The solid curve shows $C_{\mathrm{all}}(t)$, obtained by averaging the all-pairs coherence map over the selected scale band. The dashed horizontal line denotes the alarm threshold, $\theta = 0.850$, and shaded regions indicate time intervals for which $C_{\mathrm{all}}(t) > \theta$. The sustained above-threshold interval near the end of the threat window indicates persistent multiscale coherence across the distributed sensor set.

The corresponding two-dimensional coherence map is shown in Fig. 5.8. This map displays the all-pairs coherence $M_{\mathrm{all}}(a, t)$ as a function of wavelet scale and time. The strongest coherence occurred at approximately $t = 16.80$ h and scale $a = 27$, corresponding to a broad multiscale feature with an effective duration of about 13.5 minutes. Thus, while direct time-domain correlation did not recover a consistent many-body timing graph, the wavelet-domain representation exposed a coherent time-scale structure shared across the sensor ensemble. This indicates that the threat signatures successfully obscured simple localized and waveform-based indicators, but did not fully eliminate coordinated multiscale structure.

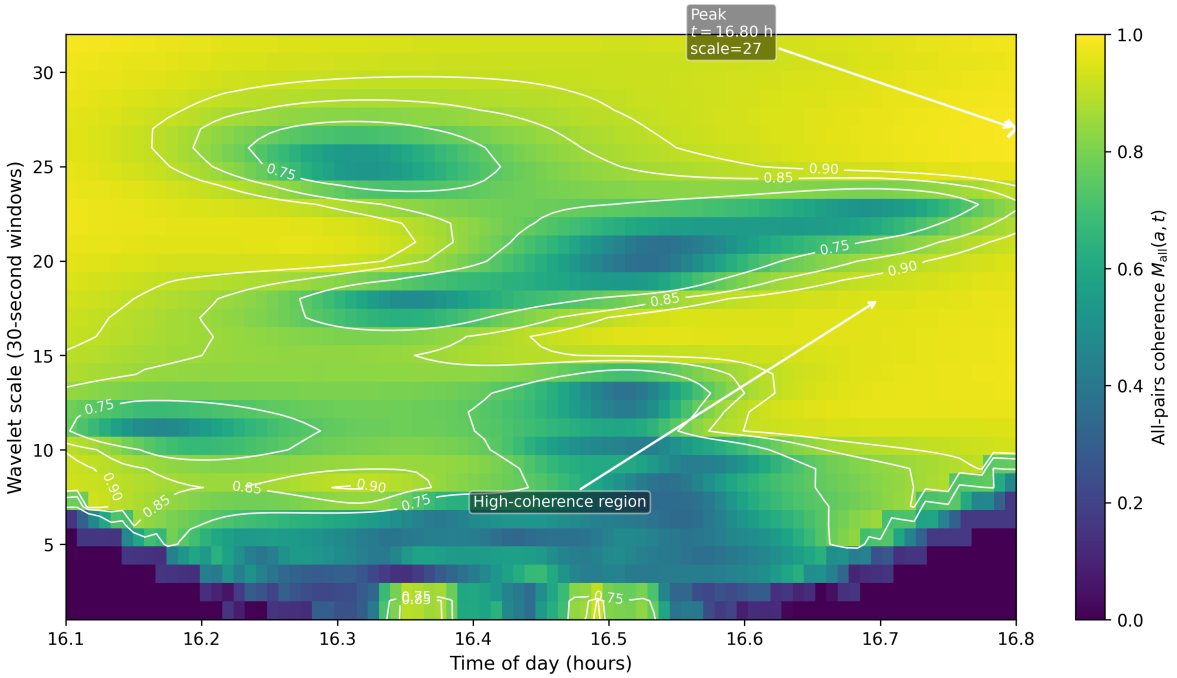


Figure 5.8: All-pairs wavelet-coherence map for the liquid-skeleton threat telemetry. The map shows $M_{\text{all}}(a, t)$, where larger values indicate stronger distributed coherence at wavelet scale a and time t. Contours highlight high-coherence regions, and the annotated peak occurs at approximately $t = 16.80$ h and scale $a = 27$, corresponding to a broad multiscale coordination structure across the sensor ensemble.

5.3 Conclusion

This project developed and evaluated a distributed network-security monitoring framework for identifying structured exfiltration-like behavior across multiple sensors. The system combined controlled telemetry replay, synthetic baseline generation, injected threat signatures, and layered mathematical detection models. The replay experiments demonstrated that the home-lab framework could reproduce both baseline-only and baseline-plus-threat telemetry with sufficient fidelity to support subsequent threat-model evaluation. Although small systematic TX/RX replay biases remained, they were consistent with expected real-system effects and did not materially undermine the detection experiments.

The mathematical analysis showed that no single detection layer is sufficient for all threat geometries. The localized attack-likelihood model identified portions of the replayed threat activity, but its response remained weak when the injected signatures were small relative to baseline traffic or partially masked by replay bias. In contrast, the lagged threat-signature correlation model successfully recovered distributed timing structure in both replayed and

synthetic exfiltration scenarios, including nonuniform Fibonacci-lag timing and mixed-signal attacks with weak, bidirectional, and geometrically dissimilar local signatures. Finally, the wavelet-coherence model detected coordinated multiscale structure in a deliberately adversarial liquid-skeleton threat signature that prevented both the localized likelihood and fixed-lag correlation models from raising alarms.

Taken together, these results support the central premise of the project: distributed exfiltration may be difficult to detect from isolated point-in-time anomalies, but coordinated structure can remain visible across sensors, time shifts, and scale-dependent representations. The layered detection architecture therefore provides a useful mathematical framework for analyzing distributed threat signatures in controlled telemetry. While the implementation is not intended to be an enterprise-grade anomaly-detection platform, it demonstrates how a modest home-lab environment can be used to construct, replay, and analyze realistic network-security scenarios with rigorous mathematical tooling. Future work could extend the framework by incorporating longer-duration low-and-slow exfiltration, richer baseline models, live packet-derived telemetry, adaptive thresholding, and more diverse multiscale detection methods.

A. Baseline Construction Parameters

Parameter	**Description**	**Value**	**Units / Notes**
Δt	Window width	30	seconds
d	Drift coefficient	0.15	Scales $B(t)P(t)$
k	Jitter coefficient	0.075	Scales $Y_{\text{smooth}}(t)Z(t)$
σ	Smoothing width	15	windows
λ	Human burst rate	20	events/day
$D_{\min}$	Minimum burst duration	2	seconds
$D_{\max}$	Maximum burst duration	10	seconds
$\mu_{\log,A}$	Burst log-normal location	$\ln(200000)$	For M_j
$\sigma_{\log,A}$	Burst log-normal scale	0.4	For M_j
n_{classes}	Operating classes	9	Workday schedule classes

Table A.1: Parameter choices used for synthetic baseline construction.

B. Replay Telemetry Parameters

Replay Parameter	Value	Description
Replay window width	30 s	Target rates were updated once per telemetry window.
Packet generation	`nc`	Persistent TCP streams supplied replay traffic.
Rate control	`tc`	Linux traffic control imposed target rates on replay TCP streams.
Queueing discipline	HTB	Hierarchical Token Bucket classes were used for rate shaping.
HTB rate-to-quantum	`tc_r2q = 100`	The rate-to-quantum parameter was increased during testing to evaluate whether HTB quantum warnings contributed to the observed TX replay bias.
HTB default class	None	No default shaped class was used, so unrelated SSH, MPI control, and NFS traffic were not restricted.
TX replay direction	Worker → injector	Worker nodes opened TCP streams to the injector on configured TX replay ports.
RX replay direction	Injector → worker	Injector opened TCP streams back to workers on configured RX replay ports.
Measurement source	/sys/.../eth0/statistics	Interface byte counters were differenced across windows to measure replayed traffic.

Table B.1: Replay telemetry configuration used for controlled replay of the baseline segment.

C. Threat Detection Analysis: Replay Telemetry

This appendix provides the complete output from the threat-detection analysis of the replay telemetry distributed exfiltration scenario. The output includes the localized and system-wide likelihood results, the per-sensor target threat intervals and volumes, the lagged residual-correlation matrices, the recovered lag matrix, and the many-body coordination alarm summary.

```
Read config file: threat_detection_config.nml
Telemetry metadata-derived dimensions:
n_sensors             =            4
n_windows             =           31
window_seconds        =           30
baseline_start_window =         1796

============================================================
Threat detection summary
============================================================
Telemetry input  : replay_target_telemetry.csv
Baseline input   : baseline.csv
Output file      : threat_detection.csv
Global output    : threat_detection_global.csv
Sensors          :            4
Windows          :           31
Window seconds   :           30
Baseline start   : window         1796 , clock 14:57:30
Observed traffic : measured replay telemetry
Expected traffic : baseline segment

Likelihood tiers :
No indication  : [0.000,  0.250)
Low            : [ 0.250,  0.500)
Moderate       : [ 0.500,  0.750)
High           : [ 0.750, 1.000]

System-wide conclusion:
Peak tier               : LOW likelihood
Peak P_sys(t)           : replay window 14, clock 15:04:00, value   0.497009
Peak P_avg(t)           : replay window 14, clock 15:04:00, value   0.146389
Detection replay windows: 10--26
Detection clock range   : 15:02:00--15:10:30
Detection duration      :    510.0 s /     8.50 min
```

```
Per-sensor conclusions:

Sensor 1: LOW likelihood
Peak P_i(t)          : replay window 13, clock 15:03:30, value   0.344841
Detection windows    : 11--14
Detection clock range : 15:02:30--15:04:30
Detection duration    :    120.0 s /     2.00 min
Target threat windows : 6--14
Target threat clock   : 15:00:00--15:04:30
Max components        : time=   0.117, magnitude=   0.212, persistence=   0.987, direction=   0.100
Target threat volume  : TX MiB=   429.2, RX MiB=     0.0

Sensor 2: LOW likelihood
Peak P_i(t)          : replay window 17, clock 15:05:30, value   0.400648
Detection windows    : 14--18
Detection clock range : 15:04:00--15:06:30
Detection duration    :    150.0 s /     2.50 min
Target threat windows : 10--18
Target threat clock   : 15:02:00--15:06:30
Max components        : time=   0.113, magnitude=   0.397, persistence=   0.987, direction=   0.160
Target threat volume  : TX MiB=   429.2, RX MiB=     0.0

Sensor 3: NO indication
Peak P_i(t)          : replay window 21, clock 15:07:30, value   0.019124
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 14--22
Target threat clock   : 15:04:00--15:08:30
Max components        : time=   0.024, magnitude=   0.035, persistence=   0.000, direction=   0.046
Target threat volume  : TX MiB=   429.2, RX MiB=     0.0

Sensor 4: LOW likelihood
Peak P_i(t)          : replay window 25, clock 15:09:30, value   0.400461
Detection windows    : 22--26
Detection clock range : 15:08:00--15:10:30
Detection duration    :    150.0 s /     2.50 min
Target threat windows : 18--26
Target threat clock   : 15:06:00--15:10:30
Max components        : time=   0.172, magnitude=   0.341, persistence=   0.987, direction=   0.165
Target threat volume  : TX MiB=   429.2, RX MiB=     0.0

Residual correlation summary:
Direction        : tx
Window source    : target_threat
Max lag windows  :           30
Lag mode         : blind scan over all allowed lags
Normalization    : z-score on full residual window; mask/zero-pad after normalization
fixed-N and support-normalized correlations both reported

Peak support-normalized correlation matrix:
Rows = sensor_i, columns = sensor_j
1.0000    0.9993    0.9925    0.9994
0.9993    1.0000    0.9951    0.9998
0.9925    0.9951    1.0000    0.9952
```

```
0.9994    0.9998    0.9952    1.0000

Peak internal lag matrix for support-normalized correlation:
Rows = sensor_i, columns = sensor_j
0         4         8        12
-4         0         4         8
-8        -4         0         4
-12        -8        -4         0

Peak fixed-N correlation matrix:
Rows = sensor_i, columns = sensor_j
0.7281    0.7283    0.7211    0.7284
0.7283    0.7296    0.7237    0.7295
0.7211    0.7237    0.7249    0.7237
0.7284    0.7295    0.7237    0.7296

Pair peak summary:
sensor_i sensor_j peak_k_support peak_plot_lag support_corr fixed_N_at_support_peak peak_k_fixed_N
↪  peak_plot_lag_fixed_N fixed_N_corr support_at_fixed_N_peak
1        1              0             0     1.000000                0.728124              0
↪  0     0.728124               1.000000
1        2              4            -4     0.999268                0.728327              4
↪  -4     0.728327               0.999268
1        3              8            -8     0.992516                0.721058              8
↪  -8     0.721058               0.992516
1        4             12           -12     0.999396                0.728403             12
↪  -12     0.728403               0.999396
2        1             -4             4     0.999268                0.728327             -4
↪  4     0.728327               0.999268
2        2              0             0     1.000000                0.729599              0
↪  0     0.729599               1.000000
2        3              4            -4     0.995113                0.723676              4
↪  -4     0.723676               0.995113
2        4              8            -8     0.999844                0.729467              8
↪  -8     0.729467               0.999844
3        1             -8             8     0.992516                0.721058             -8
↪  8     0.721058               0.992516
3        2             -4             4     0.995113                0.723676             -4
↪  4     0.723676               0.995113
3        3              0             0     1.000000                0.724870              0
↪  0     0.724870               1.000000
3        4              4            -4     0.995196                0.723719              4
↪  -4     0.723719               0.995196
4        1            -12            12     0.999396                0.728403            -12
↪  12     0.728403               0.999396
4        2             -8             8     0.999844                0.729467             -8
↪  8     0.729467               0.999844
4        3             -4             4     0.995196                0.723719             -4
↪  4     0.723719               0.995196
4        4              0             0     1.000000                0.729563              0
↪  0     0.729563               1.000000

Many-body coordination alarm:
Alarm Status                 : Alarm
Correlation threshold        :    0.700
```

```
Minimum strong edges         :        3
Lag tolerance windows        :        1
Minimum consistency fraction :    0.500
Strong unordered edges       :        6
Checked timing triplets      :        4
Consistent timing triplets   :        4
Consistency fraction         :    1.000
Interpretation               : coordinated residual timing matrix detected

NOTE: For lagged correlations, C_ij(k)=C_ji(-k).
Therefore nonzero-lag matrices are directional, not symmetric.

============================================================

Wrote detection file          : threat_detection.csv
Wrote global detection file   : threat_detection_global.csv
Wrote correlation file        : threat_detection_correlation.csv
Wrote correlation peak matrix file : threat_detection_correlation_matrix.csv
Wrote results log             : threat_detection_results.txt
```

D. Threat Detection Analysis: Fibonacci Telemetry

This appendix provides the complete output from the threat-detection analysis of the Fibonacci distributed exfiltration scenario. The output includes the localized and system-wide likelihood results, the per-sensor target threat intervals and volumes, the lagged residual-correlation matrices, the recovered lag matrix, and the many-body coordination alarm summary.

```
Read config file: threat_detection_config.nml
Telemetry metadata-derived dimensions:
n_sensors             =            4
n_windows             =         2880
window_seconds        =           30
baseline_start_window =            1

============================================================
Threat detection summary
============================================================
Telemetry input  : fibonacci_lag_exfiltration.csv
Baseline input   : baseline.csv
Output file      : threat_detection.csv
Global output    : threat_detection_global.csv
Sensors          :            4
Windows          :         2880
Window seconds   :           30
Baseline start   : window           1 , clock 00:00:00
Observed traffic : measured replay telemetry
Expected traffic : baseline segment

Likelihood tiers :
No indication  : [0.000,  0.250)
Low            : [ 0.250,  0.500)
Moderate       : [ 0.500,  0.750)
High           : [ 0.750, 1.000]

System-wide conclusion:
Peak tier               : LOW likelihood
Peak P_sys(t)           : replay window 1317, clock 10:58:00, value   0.418500
Peak P_avg(t)           : replay window 1317, clock 10:58:00, value   0.117600
Detection replay windows: 1306--1335
Detection clock range   : 10:52:30--11:07:30
Detection duration      :    900.0 s /    15.00 min
```

```
Per-sensor conclusions:

Sensor 1: LOW likelihood
Peak P_i(t)          : replay window 1309, clock 10:54:00, value   0.301807
Detection windows    : 1306--1309
Detection clock range : 10:52:30--10:54:30
Detection duration    :    120.0 s /     2.00 min
Target threat windows : 1300--1309
Target threat clock   : 10:49:30--10:54:30
Max components        : time=   0.054, magnitude=   0.106, persistence=   0.995, direction=   0.052
Target threat volume  : TX MiB=   160.2, RX MiB=     0.0

Sensor 2: LOW likelihood
Peak P_i(t)          : replay window 1317, clock 10:58:00, value   0.293661
Detection windows    : 1314--1317
Detection clock range : 10:56:30--10:58:30
Detection duration    :    120.0 s /     2.00 min
Target threat windows : 1308--1317
Target threat clock   : 10:53:30--10:58:30
Max components        : time=   0.024, magnitude=   0.108, persistence=   0.995, direction=   0.036
Target threat volume  : TX MiB=    93.4, RX MiB=     0.0

Sensor 3: LOW likelihood
Peak P_i(t)          : replay window 1322, clock 11:00:30, value   0.308152
Detection windows    : 1318--1322
Detection clock range : 10:58:30--11:01:00
Detection duration    :    150.0 s /     2.50 min
Target threat windows : 1313--1322
Target threat clock   : 10:56:00--11:01:00
Max components        : time=   0.068, magnitude=   0.113, persistence=   0.995, direction=   0.053
Target threat volume  : TX MiB=   403.6, RX MiB=     0.0

Sensor 4: LOW likelihood
Peak P_i(t)          : replay window 1335, clock 11:07:00, value   0.299831
Detection windows    : 1332--1335
Detection clock range : 11:05:30--11:07:30
Detection duration    :    120.0 s /     2.00 min
Target threat windows : 1326--1335
Target threat clock   : 11:02:30--11:07:30
Max components        : time=   0.046, magnitude=   0.106, persistence=   0.995, direction=   0.051
Target threat volume  : TX MiB=    92.9, RX MiB=     0.0

Residual correlation summary:
Direction        : tx
Window source    : target_threat
Max lag windows  :           30
Lag mode         : blind scan over all allowed lags
Normalization    : z-score on full residual window; mask/zero-pad after normalization
fixed-N and support-normalized correlations both reported

Peak support-normalized correlation matrix:
Rows = sensor_i, columns = sensor_j
1.0000    1.0000    1.0000    1.0000
1.0000    1.0000    1.0000    1.0000
1.0000    1.0000    1.0000    1.0000
```

```
1.0000    1.0000    1.0000    1.0000

Peak internal lag matrix for support-normalized correlation:
Rows = sensor_i, columns = sensor_j
0         8        13        26
-8         0         5        18
-13        -5         0        13
-26       -18       -13         0

Peak fixed-N correlation matrix:
Rows = sensor_i, columns = sensor_j
0.9965    0.9965    0.9965    0.9965
0.9965    0.9965    0.9965    0.9965
0.9965    0.9965    0.9965    0.9965
0.9965    0.9965    0.9965    0.9965

Pair peak summary:
sensor_i sensor_j peak_k_support peak_plot_lag support_corr fixed_N_at_support_peak peak_k_fixed_N
↪  peak_plot_lag_fixed_N fixed_N_corr support_at_fixed_N_peak
1        1              0             0     1.000000                0.996528              0
↪  0     0.996528               1.000000
1        2              8            -8     1.000000                0.996528              8
↪  -8     0.996528               1.000000
1        3             13           -13     1.000000                0.996528             13
↪  -13     0.996528               1.000000
1        4             26           -26     1.000000                0.996528             26
↪  -26     0.996528               1.000000
2        1             -8             8     1.000000                0.996528             -8
↪  8     0.996528               1.000000
2        2              0             0     1.000000                0.996528              0
↪  0     0.996528               1.000000
2        3              5            -5     1.000000                0.996528              5
↪  -5     0.996528               1.000000
2        4             18           -18     1.000000                0.996528             18
↪  -18     0.996528               1.000000
3        1            -13            13     1.000000                0.996528            -13
↪  13     0.996528               1.000000
3        2             -5             5     1.000000                0.996528             -5
↪  5     0.996528               1.000000
3        3              0             0     1.000000                0.996528              0
↪  0     0.996528               1.000000
3        4             13           -13     1.000000                0.996528             13
↪  -13     0.996528               1.000000
4        1            -26            26     1.000000                0.996528            -26
↪  26     0.996528               1.000000
4        2            -18            18     1.000000                0.996528            -18
↪  18     0.996528               1.000000
4        3            -13            13     1.000000                0.996528            -13
↪  13     0.996528               1.000000
4        4              0             0     1.000000                0.996528              0
↪  0     0.996528               1.000000

Many-body coordination alarm:
Alarm Status                 : Alarm
Correlation threshold        :    0.700
```

```
Minimum strong edges         :        3
Lag tolerance windows        :        1
Minimum consistency fraction :    0.500
Strong unordered edges       :        6
Checked timing triplets      :        4
Consistent timing triplets   :        4
Consistency fraction         :    1.000
Interpretation               : coordinated residual timing matrix detected

NOTE: For lagged correlations, C_ij(k)=C_ji(-k).
Therefore nonzero-lag matrices are directional, not symmetric.

============================================================

Wrote detection file          : threat_detection.csv
Wrote global detection file   : threat_detection_global.csv
Wrote correlation file        : threat_detection_correlation.csv
Wrote correlation peak matrix file : threat_detection_correlation_matrix.csv
Wrote results log             : threat_detection_results.txt
```

E. Threat Detection Analysis: Mixed-Signal Exfiltration

This appendix provides the complete output from the threat-detection analysis of the mixed-signal distributed exfiltration scenario. The output includes the localized and system-wide likelihood results, the per-sensor target threat intervals and volumes, the lagged residual-correlation matrices, the recovered lag matrix, and the many-body coordination alarm summary.

```
Read config file: threat_detection_config.nml
Telemetry metadata-derived dimensions:
n_sensors             =            4
n_windows             =         2880
window_seconds        =           30
baseline_start_window =            1

============================================================
Threat detection summary
============================================================
Telemetry input  : mixed_signal_exfiltration.csv
Baseline input   : baseline.csv
Output file      : threat_detection.csv
Global output    : threat_detection_global.csv
Sensors          :            4
Windows          :         2880
Window seconds   :           30
Baseline start   : window            1 , clock 00:00:00
Observed traffic : measured replay telemetry
Expected traffic : baseline segment

Likelihood tiers :
No indication  : [0.000,  0.250)
Low            : [ 0.250,  0.500)
Moderate       : [ 0.500,  0.750)
High           : [ 0.750, 1.000]

System-wide conclusion:
Peak tier               : NO indication
Peak P_sys(t)           : replay window 1608, clock 13:23:30, value   0.237872
Peak P_avg(t)           : replay window 1608, clock 13:23:30, value   0.060602
```

```
Detection replay windows: none
Detection clock range   : none
Detection duration      : 0 s / 0.00 min

Per-sensor conclusions:

Sensor 1: NO indication
Peak P_i(t)          : replay window 1608, clock 13:23:30, value   0.221969
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 1601--1611
Target threat clock   : 13:20:00--13:25:30
Max components        : time=   0.065, magnitude=   0.069, persistence=   0.762, direction=   0.006
Target threat volume  : TX MiB=    81.2, RX MiB=    90.9

Sensor 2: NO indication
Peak P_i(t)          : replay window 1610, clock 13:24:30, value   0.027933
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 1604--1610
Target threat clock   : 13:21:30--13:25:00
Max components        : time=   0.046, magnitude=   0.047, persistence=   0.000, direction=   0.003
Target threat volume  : TX MiB=    23.1, RX MiB=    63.7

Sensor 3: NO indication
Peak P_i(t)          : replay window 1620, clock 13:29:30, value   0.037132
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 1614--1629
Target threat clock   : 13:26:30--13:34:30
Max components        : time=   0.055, magnitude=   0.065, persistence=   0.000, direction=   0.004
Target threat volume  : TX MiB=   361.4, RX MiB=   115.3

Sensor 4: NO indication
Peak P_i(t)          : replay window 1618, clock 13:28:30, value   0.056146
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 1615--1619
Target threat clock   : 13:27:00--13:29:30
Max components        : time=   0.090, magnitude=   0.095, persistence=   0.000, direction=   0.004
Target threat volume  : TX MiB=    33.9, RX MiB=    35.0

Residual correlation summary:
Direction        : tx
Window source    : target_threat
Max lag windows  :           30
Lag mode         : blind scan over all allowed lags
Normalization    : z-score on full residual window; mask/zero-pad after normalization
fixed-N and support-normalized correlations both reported

Peak support-normalized correlation matrix:
Rows = sensor_i, columns = sensor_j
```

```
1.0000    0.8871    0.9499    0.7921
0.8871    1.0000    0.8082    0.8814
0.9499    0.8082    1.0000    0.7016
0.7921    0.8814    0.7016    1.0000

Peak internal lag matrix for support-normalized correlation:
Rows = sensor_i, columns = sensor_j
0         2        16        11
-2         0        14         9
-16       -14         0        -4
-11        -9         4         0

Peak fixed-N correlation matrix:
Rows = sensor_i, columns = sensor_j
0.9966    0.8846    0.9460    0.7901
0.8846    0.9978    0.8054    0.8797
0.9460    0.8054    0.9952    0.6994
0.7901    0.8797    0.6994    0.9984

Pair peak summary:
sensor_i sensor_j peak_k_support peak_plot_lag support_corr fixed_N_at_support_peak peak_k_fixed_N
↪  peak_plot_lag_fixed_N fixed_N_corr support_at_fixed_N_peak
1        1              0             0     1.000000                0.996577              0
↪  0     0.996577               1.000000
1        2              2            -2     0.887114                0.884623              2
↪  -2     0.884623               0.887114
1        3             16           -16     0.949908                0.946024             16
↪  -16     0.946024               0.949908
1        4             11           -11     0.792083                0.790104             11
↪  -11     0.790104               0.792083
2        1             -2             2     0.887114                0.884623             -2
↪  2     0.884623               0.887114
2        2              0             0     1.000000                0.997808              0
↪  0     0.997808               1.000000
2        3             14           -14     0.808176                0.805368             14
↪  -14     0.805368               0.808176
2        4              9            -9     0.881355                0.879696              9
↪  -9     0.879696               0.881355
3        1            -16            16     0.949908                0.946024            -16
↪  16     0.946024               0.949908
3        2            -14            14     0.808176                0.805368            -14
↪  14     0.805368               0.808176
3        3              0             0     1.000000                0.995246              0
↪  0     0.995246               1.000000
3        4             -4             4     0.701588                0.699368             -4
↪  4     0.699368               0.701588
4        1            -11            11     0.792083                0.790104            -11
↪  11     0.790104               0.792083
4        2             -9             9     0.881355                0.879696             -9
↪  9     0.879696               0.881355
4        3              4            -4     0.701588                0.699368              4
↪  -4     0.699368               0.701588
4        4              0             0     1.000000                0.998427              0
↪  0     0.998427               1.000000
```

```
Many-body coordination alarm:
Alarm Status                 : Alarm
Correlation threshold        :    0.700
Minimum strong edges         :        3
Lag tolerance windows        :        1
Minimum consistency fraction :    0.500
Strong unordered edges       :        6
Checked timing triplets      :        4
Consistent timing triplets   :        4
Consistency fraction         :    1.000
Interpretation               : coordinated residual timing matrix detected

NOTE: For lagged correlations, C_ij(k)=C_ji(-k).
Therefore nonzero-lag matrices are directional, not symmetric.

============================================================

Wrote detection file          : threat_detection.csv
Wrote global detection file   : threat_detection_global.csv
Wrote correlation file        : threat_detection_correlation.csv
Wrote correlation peak matrix file : threat_detection_correlation_matrix.csv
Wrote results log             : threat_detection_results.txt
```

F. Threat Detection Analysis: Wavelet Exfiltration - Lagged Correlation

This appendix provides the complete output from the lagged correlation threat-detection analysis of the wavelet exfiltration scenario. The output includes the localized and system-wide likelihood results, the per-sensor target threat intervals and volumes, the lagged residual-correlation matrices, the recovered lag matrix, and the many-body coordination alarm summary.

```
Read config file: threat_detection_config.nml
Telemetry metadata-derived dimensions:
n_sensors             =            4
n_windows             =         2880
window_seconds        =           30
baseline_start_window =            1

============================================================
Threat detection summary
============================================================
Telemetry input  : threat_signature.csv
Baseline input   : baseline.csv
Output file      : threat_detection.csv
Global output    : threat_detection_global.csv
Sensors          :            4
Windows          :         2880
Window seconds   :           30
Baseline start   : window            1 , clock 00:00:00
Observed traffic : measured replay telemetry
Expected traffic : baseline segment

Likelihood tiers :
No indication  : [0.000,  0.250)
Low            : [ 0.250,  0.500)
Moderate       : [ 0.500,  0.750)
High           : [ 0.750, 1.000]

System-wide conclusion:
Peak tier               : NO indication
Peak P_sys(t)           : replay window 1965, clock 16:22:00, value   0.045767
Peak P_avg(t)           : replay window 1965, clock 16:22:00, value   0.011638
```

```
Detection replay windows: none
Detection clock range   : none
Detection duration      : 0 s / 0.00 min

Per-sensor conclusions:

Sensor 1: NO indication
Peak P_i(t)          : replay window 1970, clock 16:24:30, value   0.011058
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 1951--1989
Target threat clock   : 16:15:00--16:34:30
Max components        : time=   0.018, magnitude=   0.018, persistence=   0.000, direction=   0.000
Target threat volume  : TX MiB=    59.0, RX MiB=    56.6

Sensor 2: NO indication
Peak P_i(t)          : replay window 1956, clock 16:17:30, value   0.020878
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 1955--1994
Target threat clock   : 16:17:00--16:37:00
Max components        : time=   0.035, magnitude=   0.035, persistence=   0.000, direction=   0.000
Target threat volume  : TX MiB=    69.3, RX MiB=   154.4

Sensor 3: NO indication
Peak P_i(t)          : replay window 1964, clock 16:21:30, value   0.018089
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 1963--1999
Target threat clock   : 16:21:00--16:39:30
Max components        : time=   0.030, magnitude=   0.030, persistence=   0.000, direction=   0.000
Target threat volume  : TX MiB=   119.1, RX MiB=    48.2

Sensor 4: NO indication
Peak P_i(t)          : replay window 1969, clock 16:24:00, value   0.015670
Detection windows    : none
Detection clock range : none
Detection duration    : 0 s / 0.00 min
Target threat windows : 1959--1987
Target threat clock   : 16:19:00--16:33:30
Max components        : time=   0.026, magnitude=   0.026, persistence=   0.000, direction=   0.000
Target threat volume  : TX MiB=    37.0, RX MiB=    44.8

Residual correlation summary:
Direction        : tx
Window source    : target_threat
Max lag windows  :           30
Lag mode         : blind scan over all allowed lags
Normalization    : z-score on full residual window; mask/zero-pad after normalization
fixed-N and support-normalized correlations both reported

Peak support-normalized correlation matrix:
Rows = sensor_i, columns = sensor_j
```

```
1.0000    0.8044    0.5643    0.7453
0.8044    1.0000    0.7093    0.6754
0.5643    0.7093    1.0000    0.5975
0.7453    0.6754    0.5975    1.0000

Peak internal lag matrix for support-normalized correlation:
Rows = sensor_i, columns = sensor_j
0         2         1         6
-2         0         8         4
-1        -8         0         2
-6        -4        -2         0

Peak fixed-N correlation matrix:
Rows = sensor_i, columns = sensor_j
0.9921    0.7978    0.5609    0.7396
0.7978    0.9917    0.7049    0.6701
0.5609    0.7049    0.9960    0.5941
0.7396    0.6701    0.5941    0.9926

Pair peak summary:
sensor_i sensor_j peak_k_support peak_plot_lag support_corr fixed_N_at_support_peak peak_k_fixed_N
↪  peak_plot_lag_fixed_N fixed_N_corr support_at_fixed_N_peak
1        1              0             0     1.000000                0.992068              0
↪  0     0.992068               1.000000
1        2              2            -2     0.804371                0.797830              2
↪  -2     0.797830               0.804371
1        3              1            -1     0.564294                0.560920              1
↪  -1     0.560920               0.564294
1        4              6            -6     0.745290                0.739583              6
↪  -6     0.739583               0.745290
2        1             -2             2     0.804371                0.797830             -2
↪  2     0.797830               0.804371
2        2              0             0     1.000000                0.991669              0
↪  0     0.991669               1.000000
2        3              8            -8     0.709255                0.704873              8
↪  -8     0.704873               0.709255
2        4              4            -4     0.675396                0.670090              4
↪  -4     0.670090               0.675396
3        1             -1             1     0.564294                0.560920             -1
↪  1     0.560920               0.564294
3        2             -8             8     0.709255                0.704873             -8
↪  8     0.704873               0.709255
3        3              0             0     1.000000                0.995978              0
↪  0     0.995978               1.000000
3        4              2            -2     0.597472                0.594064              2
↪  -2     0.594064               0.597472
4        1             -6             6     0.745290                0.739583             -6
↪  6     0.739583               0.745290
4        2             -4             4     0.675396                0.670090             -4
↪  4     0.670090               0.675396
4        3             -2             2     0.597472                0.594064             -2
↪  2     0.594064               0.597472
4        4              0             0     1.000000                0.992617              0
↪  0     0.992617               1.000000
```

```
Many-body coordination alarm:
Alarm Status                 : No-Alarm
Correlation threshold        :    0.700
Minimum strong edges         :        3
Lag tolerance windows        :        1
Minimum consistency fraction :    0.500
Strong unordered edges       :        3
Checked timing triplets      :        0
Consistent timing triplets   :        0
Consistency fraction         :    0.000
Interpretation               : no many-body coordination alarm

NOTE: For lagged correlations, C_ij(k)=C_ji(-k).
Therefore nonzero-lag matrices are directional, not symmetric.

============================================================

Wrote detection file          : threat_detection.csv
Wrote global detection file   : threat_detection_global.csv
Wrote correlation file        : threat_detection_correlation.csv
Wrote correlation peak matrix file : threat_detection_correlation_matrix.csv
Wrote results log             : threat_detection_results.txt


 Wavelet-coherence analysis summary
       ==================================

        Signal: sqrt(TX_threat^2 + RX_threat^2), converted to Mbps
        Mother wavelet: complex Morlet, w0 = 6
        Focus replay windows: 1933--2017
        Focus clock range   : 16:06:00--16:48:30
        CWT scales          : 1--32
        Scale band          : 8--20
        Threat volume       : TX MiB=   284.472, RX MiB=   303.985

        All-pairs statistic:
        M_all(a,t) = (1/number_of_pairs) * sum_{i<j} R_ij^2(a,t)
        C_all(t)   = mean over configured scale band of M_all(a,t)

        Coherence threshold theta       :   0.850000
        Peak C_all(t)                   :   0.939671
        Peak replay window              : 2011
        Peak clock                      : 16:45:00
        A_theta(real)                 :   0.740345 coherence-minutes
        D_theta(real)                 :  11.000000 minutes

        Wavelet-coherence alarm:
        Alarm Status                : Alarm
        Threshold theta             :   0.850000
        Minimum A_theta             :   0.300000 coherence-minutes
        Minimum D_theta             :   5.000000 minutes
        Interpretation              : coordinated multiscale wavelet structure detected

        Intervals with C_all(t) > theta:
        1933--1936 / 16:06:00--16:08:00
        1996--2017 / 16:37:30--16:48:30
```

Notes

[1]The author notes that an interpreted language such as Python would have provided more granular control over telemetry replay sequencing, and that the raw computational execution speed of Fortran 90 was not a strict requirement for this study. However, Fortran 90 was selected exclusively to eliminate black-box code interpretation. Using a compiled language where the author can fully inspect, audit, and verify every raw array operation and execution path ensured complete structural transparency across the project.

[2]Specifically, the sensors read interface statistics from sysfs counter files to determine interface activity. Because these files contain cumulative data relative to system uptime, successive differences in the counter values were used to determine network activity between heartbeats. Thus, the quantities plotted in Fig. 3.1 are byte deltas evaluated over each sampling interval.

[3]Baseline traffic generation experiments were conducted on the model $S_{\text{raw}}(t) = B(t)+D(t)+J(t)+E(t)$, where $J(\tau)$ is smoothed with the convolution. The results effectively suppressed the jitter at the local time scale.

[4]Here and elsewhere in the text, "real-valued" refers to the telemetry observables and final detection metrics. Certain intermediate representations, such as the Morlet wavelet coefficients used in the wavelet-coherence analysis, are complex-valued.

[5]Of course, in a threat modeling scenario addressing more sophisticated attack signatures, $q_i(t - t_0)$ can be informed by more than two anomaly components.

[6]This simplification is appropriate for the scope of a cybersecurity portfolio project. The objective is not to construct an enterprise-grade anomaly-detection platform capable of modeling all legitimate deviations from heuristic baselines, but to evaluate whether structured synthetic threat signatures can be detected in controlled telemetry.

[7]Several parameter choices were explored before selecting the values reported in Table **??**. A systematic optimization of those parameters is outside the scope of the present study, particularly because no enterprise traffic model is available here as an external optimization target.